ARTIFICIAL INTELLIGENCE AIDED BARANGAY ABAGA MANAGEMENT INFORMATION SYSTEM WITH N8N AUTOMATION AGENT

An Undergraduate

Research Paper presented to the

Faculty of College of Computer Science

Christ the King College de Maranding Inc.

Maranding, Lala, Lanao del Norte, Philippines

In Partial Fulfillment

Of the Requirements for Thesis 1

Bachelor of Science in Computer Science

By

GERA MAE Q. ANDO

JESSEL P. ATILLO

NEIL AELREHD Y. LAPUT

SEPTEMBER 2026

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CHAPTER 1

THE PROBLEM AND ITS SCOPE

Introduction

In the modern era of rapid technological change, local government units (LGUs) continue to face challenges in managing administrative tasks, record-keeping, and public service delivery. Many barangays, including Barangay Abaga, still rely on traditional manual methods to process official transactions such as issuing residency certificates, permits, and other essential documents. These outdated procedures often lead to inefficiencies, prolonged processing times, disorganized records, and a greater likelihood of errors. Manual systems are also prone to lost, misplaced, or inaccurate records, which adversely affect both barangay staff and residents. Consequently, residents experience long queues and delays, while barangay personnel struggle with redundant paperwork and time-consuming verification processes.

The absence of digital systems in Barangay Abaga reduces operational transparency, slows transaction processing, and limits the barangay's ability to provide responsive and citizen-centered public services. Without automation, essential services remain inaccessible to some residents who may lack time, mobility, or resources to physically visit the barangay office. Additionally, manual record-keeping makes it difficult to

generate real-time data for planning, monitoring, and decision-making, which are critical elements of effective governance.

According to Padilla (2024), in the study Enhancing Local Governance in Surigao del Norte through Barangay Connect: A Total Management Information System for Barangays, the adoption of a dedicated barangay-level Management Information System (MIS) significantly enhanced service delivery, record retrieval, and administrative efficiency within local governance. The research emphasized that before the system's implementation, barangay offices relied heavily on manual documentation, which caused frequent delays and human errors. Through automation, the system centralized resident information, streamlined document processing, and reduced waiting times for citizens availing of barangay services.

According to He, J. et al. (2025), in their study entitled Intelligent Governance: The AI-Driven New Paradigm of Governmental Adaptive Governance, the integration of artificial intelligence in government systems enabled a shift from traditional bureaucratic structures to adaptive, data-driven, and network-based governance models. The study stated that AI enhanced decision-making through real-time data analysis, predictive capabilities, and improved collaboration among stakeholders, resulting in more efficient, responsive, and innovative public service delivery systems.

According to Mamidala et al. (2024), in their study Towards Inclusive Governance: Examining Adoption of E-Governance in Rural Punjab, e-

governance systems significantly improved transparency, accountability, and service delivery in local communities. The study explained that digital platforms allowed citizens to access government services more conveniently, reducing the need for physical transactions. It also highlighted that e-governance promoted transparency by making information more accessible to the public.

The Artificial Intelligence Aided Barangay Abaga Management Information System with n8n Automation Agent was developed to streamline barangay operations by providing an integrated digital platform powered by AI and data systems. The system enabled efficient processing of residency certificates, permits, and other barangay-related transactions, allowing residents to request, track, and receive documents online without the need to visit the barangay office, thereby reducing inconvenience and long waiting times. AI-BAMIS also prioritized secure digital record-keeping to ensure data accuracy, prevent loss or tampering, and enhance transparency. To further improve efficiency, the system automated repetitive administrative tasks, minimized human errors, and accelerated processing times

The duration of this study was set from the second semester of the academic year 2025–2026 to the first semester of the academic year 2026–2027. This timeframe allowed sufficient time for the comprehensive research, development, testing, evaluation, and refinement of the AI-aided

system to ensure its usability, reliability, and effectiveness within the community.

This project was developed by a team of Computer Science students with expertise in software development and database management. To ensure the reliability and efficiency of AI-BAMIS, the researchers conducted interviews and surveys with key barangay officials, including the barangay captain, secretary, treasurer, and barangay councilors, who provided insights into the challenges of manual record-keeping and administrative tasks. Additionally, feedback from barangay staff and residents was gathered to assess the specific needs of the community. The research was grounded in reputable sources, existing studies on digital governance, and proven methodologies in software engineering. Through direct collaboration with local officials and stakeholders, the system was designed to address real-world issues, ensuring its effectiveness and relevance in improving barangay administrative processes.

Theoretical Framework

This chapter presented the theoretical foundations that guided the development and evaluation of the Artificial Intelligence Aided Barangay Abaga Management Information System with n8n Automation Agent. The study focused on achieving the objectives of AI-BAMIS by applying established models of technology acceptance and information system

success that assess system effectiveness, user adoption, and sustainability. The research was grounded on three main theories: the DeLone and McLean Information Systems Success Model (D&M Model), the Technology Acceptance Model (TAM), and the Unified Theory of Acceptance and Use of Technology (UTAUT).

The DeLone and McLean Information Systems Success Model (D&M Model) provides a framework for evaluating information system success through six dimensions: system quality, information quality, service quality, use, user satisfaction, and net benefits. Katili et al. (2024) applied this model to assess a Village Information System in Tabongo Village, Gorontalo Regency, and found that system quality, information quality, and user satisfaction significantly influenced net benefits.

The Technology Acceptance Model (TAM), developed by Davis, highlights that perceived usefulness and perceived ease of use are key determinants of technology adoption. This was supported by the study of Distor et al. (2021), which examined the acceptance of AI in the Municipality of Carmona, Cavite. The study applied TAM combined with UTAUT constructs to assess the perceptions of local government officials and department administrators regarding AI adoption in governance. The findings revealed that perceived usefulness and perceived ease of use significantly influenced positive attitudes and behavioral intention to use AI in government operations. The results also showed generally moderate to

high acceptance levels among respondents, indicating readiness for AI integration in local governance.

The Unified Theory of Acceptance and Use of Technology (UTAUT) integrates multiple determinants of technology adoption, including performance expectancy, effort expectancy, social influence, and facilitating conditions. In the study entitled UTAUT Model Mediated by Government Information System to the Intention to Use of Indonesian Local Government, Hasanah et al. (2024) examined the application of the UTAUT model in the implementation of the Sistem Informasi Pemerintahan Daerah (SIPD) in Indonesian local government units. The study involved 358 respondents, including high-ranking officials, administrators, supervisors, and system management personnel. Using descriptive statistical analysis and Structural Equation Modeling (SEM), the researchers found that performance expectancy, effort expectancy, social influence, and facilitating conditions all had significant effects on users' intention to use the SIPD system.

The findings emphasized that improving system usefulness, ease of use, social encouragement, and institutional facilitation was essential to ensure successful adoption and sustained use of government information systems. These concepts were highly relevant to the implementation and evaluation of the Artificial Intelligence Aided Barangay Abaga Management Information System with n8n Automation Agent.

Conceptual Framework

The researchers developed the Artificial Intelligence Aided Barangay Abaga Management Information System with n8n Automation Agent to address inefficiencies in traditional barangay operations, particularly in resident record management and document request processing. Figure 1 shows the Input-Process-Output (IPO) model of the system.

The Input included resident information, user roles, service request details, uploaded requirements, administrator credentials, and AI agent inquiries. These served as the primary data sources for system operations. The Process stage involved secure user authentication, data validation, centralized storage, and automated workflow processing. The system integrated AI-assisted validation to support request classification. Through the n8n Automation Agent, an AI-powered assistant handled user inquiries and guided residents, while barangay staff reviewed and managed requests through an administrative dashboard. The system also supported real-time status tracking and automated logging to improve transparency and efficiency. The system followed the Systems Development Life Cycle (SDLC) methodology, which ensured structured development through the phases of planning, analysis, design, implementation, testing, deployment, and maintenance to achieve reliability, security, and scalability. The Output was a fully functional Artificial Intelligence-Aided Barangay Abaga Management Information System powered by an n8n Automation Agent.

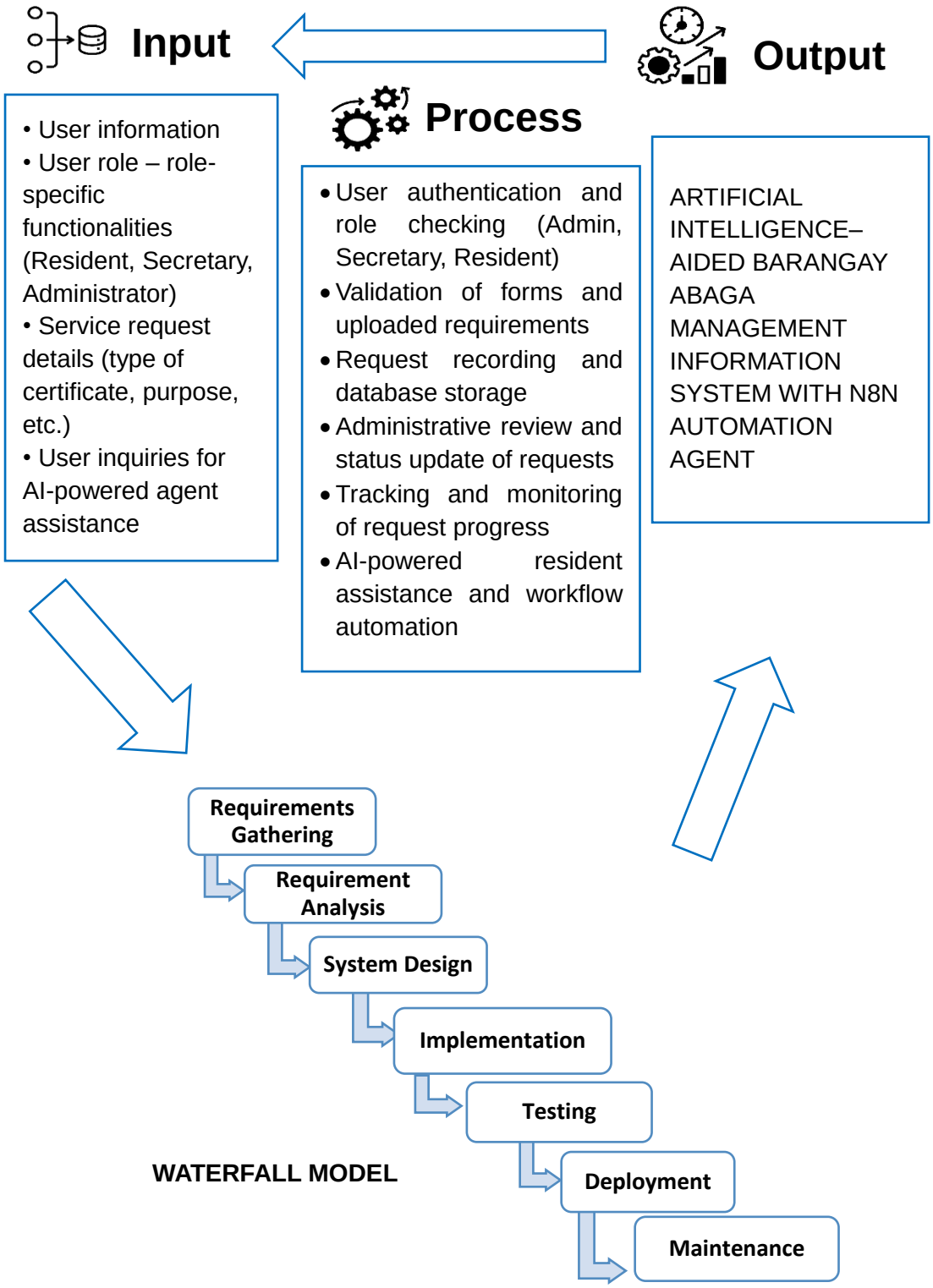


***Figure 1.*Schematic Diagram (Input-Process-Output Model)**

Statement of the Problem

Barangay Abaga relied on a manual system for processing residency certificates, indigency certificates, business permits, community tax certificates (cedula), barangay clearances, blotter requests, and other barangay-related transactions. This traditional approach resulted in inefficiencies, prolonged processing times, disorganized record-keeping, and a higher risk of human errors. As a result, residents experienced delays in accessing essential services, while barangay officials faced difficulties in monitoring records, tracking transactions, and ensuring data accuracy and transparency.

To address these concerns, this study developed the Artificial Intelligence-Aided Barangay Abaga Management Information System with n8n Automation Agent. The system aimed to streamline administrative transactions through automation and artificial intelligence integration, enhance service accessibility, strengthen data security, and provide a user-friendly digital platform for both barangay officials and residents. Specifically, this study sought to answer the following questions:

1. What were the current challenges faced by Barangay Abaga in managing resident records and processing barangay transactions under the existing manual system?
2. What technologies, methodologies, and AI components were essential in developing an effective Artificial Intelligence-Aided

Barangay Abaga Management Information System with n8n Automation Agent?

3. How could an AI-integrated automated system address the operational needs, preferences, and expectations of barangay officials and residents?
4. How could the Artificial Intelligence-Aided Barangay Abaga Management Information System with n8n Automation Agent improve overall user satisfaction among residents and barangay staff?
5. How could the Artificial Intelligence-Aided Barangay Abaga Management Information System with n8n Automation Agent be evaluated using the ISO 9126-1 Standard Evaluation Tool?

Scope and Delimitations

Scope

The Artificial Intelligence-Aided Barangay Abaga Management Information System with n8n Automation Agent was a web-based platform developed to improve the efficiency, transparency, and service delivery of barangay operations in Barangay Abaga. The system was designed to digitalize and streamline essential barangay processes, particularly those related to resident services and administrative transactions.

The system covered functionalities such as resident registration, secure user authentication, and comprehensive resident profiling. It also

supported the online submission, processing, and tracking of barangay service requests. These included Barangay Clearance, Certificate of Residency, Certificate of Indigency, Business Permit-related requests, Community Tax Certificate (Cedula), blotter-related concerns, and other authorized barangay transactions.

In addition, the system provided centralized record management, income recording and reporting, as well as the generation of administrative reports to support efficient decision-making. A key feature of the system was the integration of an AI-powered automation agent through n8n. This agent served as an active assistant that guided users through processes, validated inputs, triggered automated workflows, and assisted barangay personnel in managing and routing requests more efficiently. The system also implemented role-based access control and security features such as email verification, OTP authentication, and password recovery. Designed for internet-enabled devices, AI-BAMIS aimed to enhance accessibility and improve the overall delivery of barangay services.

Delimitations

The study was limited to a web-based environment; thus, a stable internet connection was required to access and utilize the system's full functionalities. Users without internet access were unable to use the online services provided by the system. The scope of the study was confined to

barangay-level operations within Barangay Abaga. It focused only on selected services such as resident management, online request processing, document issuance, record management, income tracking, reporting, and workflow automation.

The study did not extend to municipal, provincial, or national government systems, nor did it include payroll systems or full-scale financial accounting systems. Although the system integrated an AI-powered automation agent, its role was limited to assisting, validating, and automating parts of the workflow. Final approval, verification, and release of official barangay documents remained under the manual control of authorized barangay personnel in accordance with existing rules and procedures.

Significance of the Study

The Artificial Intelligence-Aided Barangay Abaga Management Information System with n8n Automation Agent was developed to enhance the efficiency, transparency, and intelligence of barangay administrative processes by integrating digital technologies and Artificial Intelligence (AI) into transaction handling and record management while promoting innovation in local governance systems.

This study was significant as it benefited various stakeholders, including the barangay association (client), students, teachers, school

administrators, and future researchers. It demonstrated how AI-driven digital systems could strengthen local governance, improve public service delivery, and support effective and sustainable data-driven decision-making processes.

The Barangay Association. AI-BAMIS provided barangay officials with an organized, automated, and intelligent system for managing transactions and records. Through digital record-keeping, AI-assisted request classification, and agent support, barangay staff were able to reduce paperwork, minimize processing delays, and improve overall workflow efficiency. The system enhanced transparency by maintaining accurate transaction logs and monitoring service trends. Security features such as email verification, role-based access control, and secure authentication mechanisms helped protect sensitive data and ensure that only authorized personnel could access administrative functions.

Students. The system served as a practical and innovative learning tool for students, particularly in the fields of computer science and information technology. It demonstrated how Artificial Intelligence, rule-based systems, and data analytics could be applied in real-world governance settings. Students were able to analyze the system's database structure, security mechanisms, and AI classification logic. This exposure enhanced their technical knowledge and prepared them for future careers in system development, AI integration, and digital transformation projects.

Teachers. AI-BAMIS served as an instructional example of real-world applications of information technology, artificial intelligence, e-governance, and database management. Teachers could utilize the system to illustrate how automation and intelligent systems improved administrative efficiency and public service delivery. It provided a concrete model that encouraged innovation and critical thinking in designing technology-driven solutions for community development.

School Administrators. The system highlighted how digital transformation and AI-assisted processes could improve organizational efficiency. It provided insights that could inspire school administrators to adopt similar intelligent systems for managing school transactions, document processing, and administrative workflows. By observing the impact of AI-supported automation in barangay operations, administrators could explore modernization initiatives within educational institutions, leading to more efficient, transparent, and streamlined academic management processes.

Future Researchers. This study provided a valuable reference for future researchers interested in AI integration, digital governance, and community-based information systems. It offered insights into the development, implementation, and evaluation of the Artificial Intelligence-Aided Barangay Abaga Management Information System with n8n Automation Agent. Future studies may build upon this research by enhancing AI capabilities, expanding system coverage, or applying similar models to other local

government units. The study contributed to the growing body of knowledge on AI-driven public service innovation and local governance digitalization.

Definition of Terms

Administrative Tasks. These refer to the duties performed by barangay officials and staff, including reviewing requests, managing records, and approving documents.

Automated System. It refers to the digital platform that replaces manual barangay processes through system-driven operations to improve efficiency and accuracy.

AI agent. It refers to the system feature that provides automated guidance to residents, answers frequently asked questions, and assists users in tracking the status of their document requests.

Barangay Officials. These refer to the authorized personnel, including the Barangay Captain, Secretary, Treasurer, and Admin, who manage and oversee the system's operations.

Barangay Staff. These refer to personnel assigned to assist in administrative functions, including processing document requests, managing resident information, and maintaining records.

Centralized Database. It refers to the secured digital storage system where all resident information, document requests, and transaction records are stored and managed.

Data Security. It refers to the protective measures implemented to safeguard resident information and transaction records from unauthorized access or tampering.

Digital Record-Keeping. It refers to the electronic storage and organization of barangay documents and resident data.

Document Processing. It refers to the procedure where document requests are reviewed, validated, approved, or rejected by authorized officials.

Efficiency. It refers to the improved speed and accuracy of completing barangay transactions through system-assisted automation.

Human Errors. These refer to mistakes made during manual processing, such as misfiling records, incorrect document issuance, and data inconsistencies.

Information Management. It refers to the systematic handling of barangay data, including storage, retrieval, updating, and monitoring of resident records and transactions.

n8n Workflow Automation. It refers to the backend automation system integrated into the platform that functions specifically as an AI agent assistant for the agent feature

Online Requests. These refer to document applications submitted by residents through the web-based platform.

Processing Time. It refers to the duration required for barangay officials to review, approve, and issue requested documents.

Resident Profiling. It refers to the digital registration and storage of residents' personal and demographic information.

Role-Based Access Control. It refers to the system security mechanism that restricts access to specific features based on the assigned role of the user.

Transaction Monitoring. It refers to the system's capability to track, record, and update the status of all barangay document requests and related activities.

Transparency. It refers to the system's ability to maintain accurate transaction logs and provide clear status updates to ensure accountability in barangay operations.

User Satisfaction. It refers to the level of approval and positive experience of residents and officials when using the barangay system for document processing and administrative tasks

CHAPTER 2

REVIEW OF RELATED LITERATURE AND STUDIES

This chapter summarized the key findings from related literature that were gathered by the researchers and were relevant to the investigation.

Related Literature

This section focuses on theoretical and conceptual perspectives that establish the foundation for integrating artificial intelligence and digital systems into local governance.

According to Collins et al. (2021), in their study entitled Artificial Intelligence in Information Systems Research: A Systematic Literature Review and Research Agenda, artificial intelligence significantly enhances organizational performance through process automation, data-driven insights, and improved customer engagement. The study explains that AI enables organizations to streamline repetitive tasks, reduce human error, and increase overall productivity. It further highlights that machine learning is the most widely used AI technique, allowing systems to learn from data and generate accurate predictions. Through these capabilities, organizations are able to make faster and more informed decisions, improving operational efficiency. However, the authors noted that the absence of a unified definition of AI creates inconsistencies in research and

implementation. This limitation affects how AI systems are designed and evaluated across different sectors.

Similarly, Fibriyanita (2024), in the study The Rise of Artificial Intelligence (AI) in Public Administration: Reimagining Government Services and Delivering Value to Citizens, highlighted that AI improves public service delivery by increasing efficiency, productivity, and decision-making capabilities. The study explains that AI allows government agencies to automate administrative tasks, reducing processing time and improving service accuracy. It also emphasizes that AI enhances citizen satisfaction by providing faster and more responsive services. Furthermore, the integration of AI enables governments to analyze large volumes of data, supporting evidence-based policymaking.

Consistent with these findings, Andaya et al. (2025), in Digital Governance in the Philippines: A Scoping Review of Current Challenges and Opportunities, emphasized that digital governance enhances efficiency, transparency, and citizen engagement. The study explains that digital platforms allow faster transactions and improved communication between citizens and government units. It also highlights that digital governance strengthens accountability by reducing manual processes and minimizing opportunities for corruption. However, the study identified major challenges such as inadequate ICT infrastructure, financial constraints, and resistance to technological adoption. These barriers hinder the effective

implementation of digital systems in local government units. The authors further emphasized that emerging technologies like AI offer great potential but are still underutilized.

Furthermore, Jinghua et al. (2025), in their study Intelligent Governance: The AI-Driven New Paradigm of Governmental Adaptive Governance, introduced the concept of intelligent governance, where AI enables governments to transition toward adaptive and data-driven decision-making. The study explains that AI allows real-time data analysis, enabling governments to respond quickly to changing conditions. It also highlights that AI supports flexible decision-making by providing predictive insights and automated recommendations. Additionally, the integration of AI promotes collaboration among different sectors, improving coordination and service delivery. The study emphasizes that intelligent governance systems are more resilient and capable of handling complex societal challenges.

Additionally, Aljabri (2024), in Driving Efficiency: The Role of Artificial Intelligence (AI) in Enhancing Municipal Operations in Saudi Arabia, found that AI significantly improves efficiency, resource management, and service delivery in municipal operations. The study explains that AI can be applied in various areas such as urban planning, waste management, and citizen services. It highlights that AI enables better allocation of resources by analyzing data and predicting future needs. Furthermore, AI improves decision-making by providing accurate and real-time insights. However, the

study also identified challenges such as data privacy concerns, regulatory limitations, and lack of workforce readiness. These issues hinder the full implementation of AI technologies in government systems. Therefore, the study reinforces the importance of addressing both technical and organizational challenges when integrating AI into local governance systems.

Related Studies

This section presents empirical studies that focus on the development, implementation, and evaluation of barangay management systems and related technologies.

According to Albano (2024), in his study Empowering Local Government Units with Open-Source Tools: Building a Dynamic Web-Based Information System, the development of a web-based information system significantly improved transparency and accessibility within the Local Government Unit of Cabagan, Isabela. The study emphasized the use of open-source technologies to provide public access to municipal accomplishments, financial information, and essential government services. Findings revealed that the system was highly satisfactory in terms of design, navigation, and content presentation, allowing users to easily access relevant information. The study also highlighted that digital platforms can strengthen accountability, promote citizen engagement, and enhance public

service delivery. Furthermore, the use of cost-effective open-source tools demonstrated the potential of technology in supporting efficient local governance and improving communication between government offices and constituents.

The study by Jacobe et al. (2021), in their study Barangay Profiling System with Analytics, automation of resident data management significantly improves efficiency in barangay operations. The study explains that organizing resident information digitally allows easy access and retrieval of data. It also highlights that statistical reports help officials analyze population trends and community needs. Furthermore, analytics tools support data-driven decision-making. The system reduces redundancy and improves accuracy in data handling. It also enhances planning and service delivery.

According to Balilia et al. (2023), in their study Information Management System with Project Monitoring for Barangay, the development of a computerized Barangay Information Management System significantly improved the efficiency of barangay operations. The system automated the management of resident records, document requests, session reports, and barangay projects, reducing reliance on traditional manual record-keeping methods. The study highlighted that project monitoring features enhanced transparency and accountability in managing barangay projects, while automated document generation improved service

delivery to residents. Evaluation results showed that users found the system useful, easy to use, and satisfactory .

According to Ballaran et al. (2023), in their study Development and Evaluation of Web-Based Barangay Profiling and Issuance System Using Regression Analysis, the implementation of a web-based barangay profiling and issuance system significantly improves the management of resident information and document processing. Furthermore, the system facilitated faster issuance of barangay clearances, certificates, and other essential documents while providing analytical tools for decision-making through regression analysis. The researchers found that the system achieved high ratings in functionality, usability, security, reliability, and portability, indicating strong user acceptance.

Carpio (2020), in Barangay Management System, emphasized that web-based systems improve governance by automating administrative processes. The study explains that online services allow residents to request documents conveniently. It also highlights that automation reduces processing time and errors. Furthermore, the system increases transparency by providing accessible information. It also promotes citizen participation through digital platforms.

According to Dela Merced and Pagtaconan (2022), in their study Development of an Information Systems Strategic Plan for the Municipality of Nueva Era, Ilocos Norte, the development of an Information Systems

Strategic Plan (ISSP) plays a significant role in improving the efficiency and effectiveness of local government operations. The study emphasized the importance of integrating Information and Communication Technology (ICT) into government services to streamline transactions, improve data management, and enhance public service delivery. It identified several operational challenges caused by manual processes and proposed systems such as a Barangay Resident Information System, Human Resource Management System, Business Permit and Licensing System, Payroll Processing System, and Budget and Expense Monitoring System.

Cuevas et al. (2024), in their study Enhancing the Barangay Information System by Integrating Geolocation Technology and Tagalog agent using NLP Algorithm for Efficient Community Engagement and Service, demonstrated that agent integration improves communication and accessibility. The study explains that agents provide instant responses to citizen inquiries. It also highlights that geolocation improves disaster response and service coordination. Furthermore, the system reduces workload for barangay personnel. It enhances engagement between citizens and government. It also improves response time and service efficiency.

Pastoril (2023), in Unified Barangay E-Services System with Augmented Analytics, found that augmented analytics enhances decision-making capabilities. The study explains that analytics tools provide insights

into service usage and trends. It also highlights that these insights support planning and resource allocation. Furthermore, the system improves efficiency by reducing processing time. It also enhances data organization. The findings show improved service delivery and transparency.

Maribao et al. (2024), in Enhancement of Barangay Management System, highlighted that integrated systems improve efficiency and transparency. The study explains that centralized data management reduces redundancy and errors. It also highlights that automation improves workflow and service delivery. Similarly, Aliling et al. (2025) emphasized that digital systems enhance accessibility and efficiency. These systems streamline processes and improve communication. They also support better governance practices.

Finally, Yumen (2025), in Computer Literacy for Barangay Health Workers, identified that digital skills significantly improve system utilization. The study explains that trained personnel can efficiently use digital systems for reporting and data management. However, it also highlights challenges such as lack of training and limited infrastructure. These barriers affect system adoption and effectiveness. The study emphasizes the need for continuous training programs. It also highlights the importance of improving access to technology. This underscores the need for user training and system usability in implementing BAMIS.

Insights Learned from the Related Literature and Studies

The reviewed literature and studies reveal that integrating artificial intelligence and digital technologies significantly improves efficiency, transparency, and decision-making in local governance. AI plays a crucial role in automating processes, analyzing data, and supporting intelligent decision-making. Meanwhile, digital systems enhance service delivery by improving accessibility and communication between citizens and government units. Another important insight is that modern barangay systems must incorporate advanced features such as analytics, decision support systems, cloud computing, and communication tools. These technologies enable faster, more accurate, and more responsive services. Additionally, integrating features like GIS and agents enhances system functionality and citizen engagement.

Furthermore, user-centered design and inclusivity are essential to ensure system adoption, especially in rural communities. Training programs, awareness campaigns, and infrastructure development are necessary to support digital transformation. Systems must be easy to use and accessible to all users. Overall, the integration of AI and digital technologies provides a strong foundation for developing an intelligent, efficient, and sustainable BAMIS.

CHAPTER 3

RESEARCH METHODOLOGY

The methodology of the investigation was presented in this chapter. This chapter discussed the research environment, research respondents, and research design.

Research Design

To ensure that the Artificial Intelligence-Aided Barangay Abaga Management Information System with n8n Automation Agent operated effectively and efficiently, delivering accurate results and streamlined services, the researchers utilized a well-designed system architecture integrated with AI capabilities and workflow automation. The incorporation of the n8n automation agent as an intelligent assistant enhanced system functionality by automating processes such as request handling, agent assistance, notifications, and status tracking.

The implementation of a structured and intelligent framework improved the reliability and accuracy of barangay data, enabling administrators to make informed, data-driven decisions while maintaining transparency and user satisfaction. Furthermore, the system design optimized data flow, accessibility, and real-time interaction, ensuring that residents and barangay personnel experienced a smooth and efficient process from service request submission to approval, monitoring, and completion.

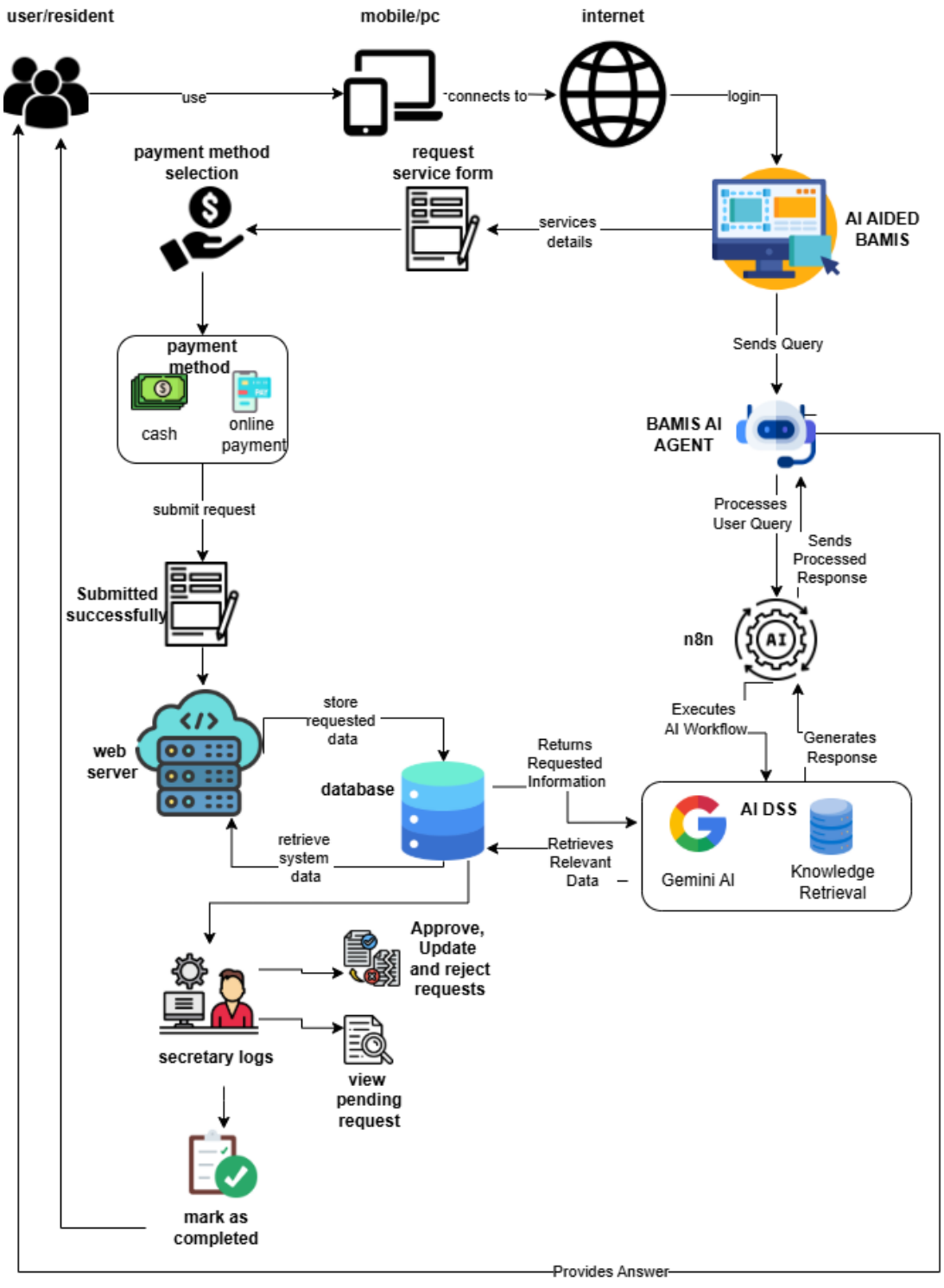


Figure 2. **Detailed Design of AI-Aided BAMIS**

Figure 2 presents the detailed design of the Artificial Intelligence-Aided Barangay Abaga Management Information System with n8n Automation Agent (AI-Aided BAMIS). It illustrates the interaction among residents, the barangay secretary, the web server, database, AI Agent, and decision support components in processing barangay service requests. The design shows how information flows through the system, from user access and request submission to approval, storage, and automated response generation.

The process began when a resident accessed the system using a mobile device or computer connected to the internet. After logging in, the resident was able to view service information, complete a request form, and select a preferred payment method, either cash or online payment. Once submitted, the request was transmitted to the web server, where the information was processed and stored in the system database. The database served as the central repository for resident records, requests, transactions, and other relevant information.

The barangay secretary accessed the administrative interface to review pending requests submitted by residents. The secretary was able to verify information, approve, update, or reject applications, and mark completed requests accordingly. This process ensured that request records remained updated and properly managed. In addition, the system provided a centralized platform for managing barangay transactions and records.

Through the integration of the web server and database, information was securely stored, retrieved, and updated whenever necessary. This ensured data consistency, minimized manual processing, and supported efficient handling of resident requests.

The system also incorporated the BAMIS AI Agent, which processed user inquiries through automated workflows powered by n8n. The AI Agent communicated with the Artificial Intelligence Decision Support System (AI DSS), utilizing Gemini AI and a Knowledge Retrieval component to retrieve relevant information from the database and generate responses. This integration supported request processing, information access, improved communication, and enhanced delivery of barangay services while providing timely assistance, reducing manual inquiries, and improving the overall user experience within the system.

Architectural Design (SDLC Waterfall Model)

The architectural design of the Artificial Intelligence–Aided Barangay Abaga Management Information System with n8n Automation Agent was developed following the SDLC Waterfall Model, which consisted of structured phases. The process began with the Requirements Gathering phase, where the researchers collected and analyzed information about the barangay's existing manual processes and identified the needs of both barangay officials and residents. This ensured that the system design aligned with operational requirements and user expectations.

This was followed by the System Design phase, where the researchers organized and structured the collected data to create a comprehensive system architecture, including the database design, user interface, and integration of AI-assisted features through the n8n automation agent. The design focused on improving the efficiency, usability, and accessibility of barangay services.

During the Coding phase, the system was developed using Visual Studio Code as the primary programming environment. Technologies such as PHP, HTML, CSS, and JavaScript were used to build the web-based interface and system functionalities, while MySQL served as the centralized database for storing resident records, service requests, financial transactions, and reports. The n8n automation agent was also integrated to support agent assistance and workflow automation.

Next, during the Testing and Debugging phase, the system underwent thorough testing to identify and resolve errors, ensuring accuracy, functionality, security, and overall system performance. All modules were evaluated to confirm that they operated according to the intended design. After successful testing, the system proceeded to the Deployment phase, where it was deployed online and made accessible through the internet. Although the system became functional and accessible, it had not yet been fully implemented in the actual barangay environment for real users.

Finally, the system entered the Maintenance and Support phase, where continuous monitoring, updates, and improvements were conducted to enhance performance, address issues, and ensure that the system remained reliable and adaptable to future user needs.

Waterfall Model

The researchers' development of the Artificial Intelligence–Aided Barangay Abaga Management Information System with n8n Automation Agent (AI-Aided BAMIS) followed the structured Waterfall Model, which allowed each phase, such as requirements gathering, system design, implementation, testing, and deployment, to build upon the previous phase. This approach ensured thorough planning, design, and testing, resulting in a robust system that aligned with the operational needs of Barangay Abaga.

A waterfall diagram illustrated the flow of tasks throughout the system development lifecycle. Although the Waterfall Model provided a structured foundation for development, selected Agile practices were incorporated to accommodate evolving requirements and feedback during the development process. This approach enabled the researchers to make necessary adjustments based on evaluation results and stakeholder feedback, contributing to a reliable, user-friendly, and functional system. The combination of structured Waterfall phases and adaptive practices supported continuous system improvement throughout development, testing, deployment, and maintenance, while enhancing the system's

usability, scalability, reliability, and potential for future integration with emerging technologies.

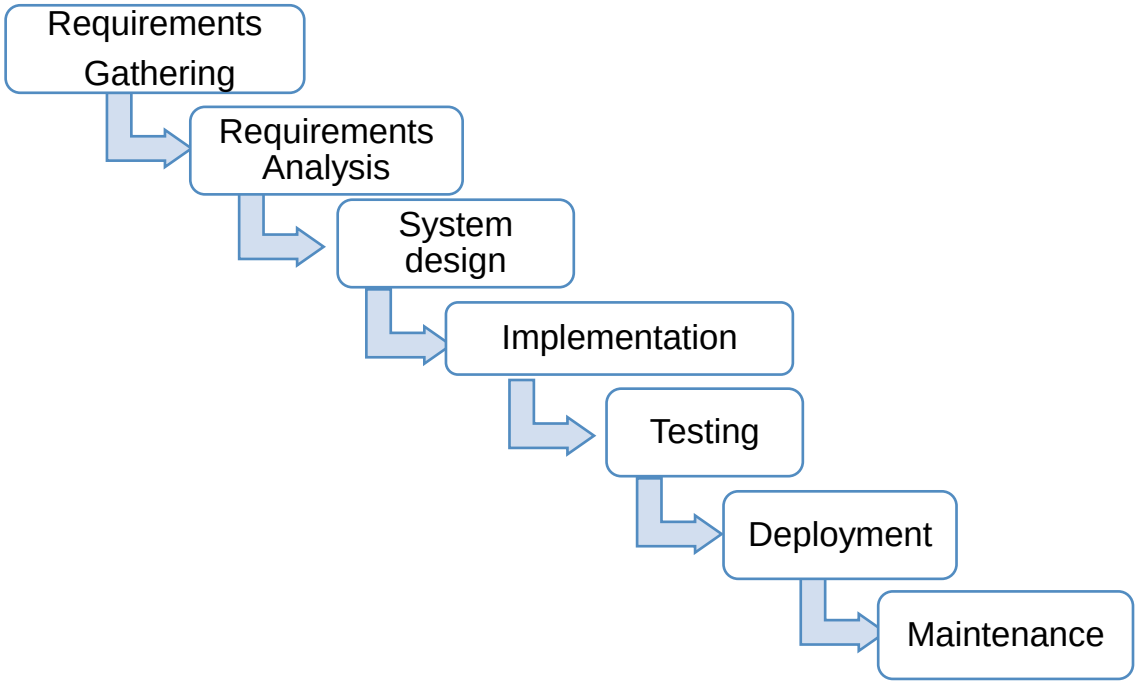


Figure 3. **Waterfall Model**

The development process using the Waterfall Model involved the following steps:

Requirements Gathering. The first phase involved collecting and documenting the system requirements by identifying the needs of barangay officials, staff, and residents. Requirements gathering was a vital process in developing the Artificial Intelligence-Aided Barangay Abaga Management Information System with n8n Automation Agent (AI-Aided BAMIS). This phase focused on identifying the specific needs of the barangay in managing records, services, and transactions efficiently, while also exploring how artificial intelligence could assist in improving these processes. During this

stage, the researchers consulted barangay officials, staff members, and residents to determine the existing problems associated with the manual or traditional processes of handling barangay services. The information gathered helped determine how AI-supported features, such as automated responses and decision-support assistance, could enhance the efficiency and accessibility of barangay services. These findings contributed to faster service delivery, improved accuracy, and a better user experience across administrative operations.

Requirement Analysis. Requirement analysis served as a fundamental phase in the development of the Artificial Intelligence-Aided Barangay Abaga Management Information System with n8n Automation Agent. The main objective of this phase was to thoroughly gather, analyze, and document the specific system requirements, including management information features and artificial intelligence components.

This process involved engaging with barangay officials, staff, and other stakeholders through interviews, surveys, and direct observations to identify their operational needs, challenges, and expectations. The collected requirements were carefully evaluated to determine their feasibility, prioritize essential features, and ensure that they were clear, complete, and aligned with the goals of the barangay. These requirements served as the foundation for the design and development of AI-Aided BAMIS, ensuring that

the system effectively supported barangay operations while incorporating AI-powered assistance.

Table 1. **Software Requirements of the AI AIDED BAMIS**

Software Requirements	Details	Description
Front-end	HTML / CSS JavaScript	Used to design and structure web pages, adding interactivity and dynamic features.
Back-end	Server-Side Language Database	Handles server-side processing, system logic, and database management for storing data.
Workflow Automation / AI Integration	n8n	Automates tasks, notifications, and integrates AI features for smoother system operations.
Integrated Development Environment (IDE)	Visual Studio Code Cursor	Provides tools for coding, debugging, and developing web applications efficiently
Web Server	XAMPP	Simulates a local server for testing and running the system before deployment.
Operating System	Windows 11	Serves as the platform that supports and runs the system smoothly.
Browser Compatibility	Google Chrome Microsoft Edge	Ensures the system works properly and displays correctly across different web browsers.

Table 1 presents the essential software components used in the development and implementation of the Artificial Intelligence-Aided Barangay Abaga Management Information System with n8n Automation Agent. The system was developed as a web-based application that served as a centralized platform for managing barangay records, handling transactions, and enhancing administrative efficiency. The front-end

technologies, including HTML, CSS, and JavaScript, were used to design responsive web pages, enable interactivity, and ensure smooth user navigation. To enhance automation and AI-assisted features, n8n was integrated into the system to manage workflows, enable automated notifications, and support the AI-powered agent for residents and barangay staff. Visual Studio Code served as the Integrated Development Environment (IDE), providing tools for writing, debugging, and managing source code.

Table 2. **Hardware Requirements of the AI AIDED BAMIS**

Hardware Requirements	Details	Description
Servers	Web Server	Processor: Quad-core or higher. RAM: 16GB or higher. Storage: SSD with at least 500GB. Network: Stable, high-speed internet connection.
	Database Server	Processor: Quad-core or higher. RAM: 32GB or higher. Storage: SSD with RAID configuration fast access. Network: High-speed and secured internet connection.
Network	Firewall VPN	Ensures protection of the system's internal network from unauthorized access.
Workstations	Developer Workstations	Processor: Quad-core or higher. RAM: 16GB or higher. Storage: SSD with at least 256GB. Operating System: Windows 10/11 or Linux

Backup and Storage	Backup Server Cloud Storage	Processor: Quad-core or higher. RAM: 16GB or higher. Storage: Large-capacity HDDs with RAID configuration. Cloud Services: Google Drive, AWS S3, or Google Cloud
Miscellaneous	Power Supply Cooling System	Power Supply: Uninterruptible Power Supply (UPS) Cooling System

Table 2 presents the necessary hardware requirements that supported the efficient operation of the Artificial Intelligence-Aided Barangay Abaga Management Information System with n8n Automation Agent. The system relied on servers for web hosting and database management to handle simultaneous user access, data processing, and secure data storage. A stable network environment was maintained through appropriate security measures to protect the system and support secure remote connections. Developer workstations with sufficient processing power and memory supported system development, updates, and maintenance. Backup mechanisms, including a dedicated backup server and cloud storage, helped safeguard data integrity against potential data loss. To support uninterrupted operations, UPS units and appropriate cooling systems were utilized as part of the barangay's IT infrastructure.

System Design. In this phase, the architectural framework and detailed specifications of the Artificial Intelligence-Aided Barangay Abaga Management Information System with n8n Automation Agent were

developed based on the analyzed requirements. The system design defined the overall structure, key components, user interfaces, and data flow within the system. This phase was essential in ensuring that AI-Aided BAMIS would effectively support the barangay's administrative operations, service delivery, and AI-assisted functions such as automated workflows and agent interactions.

It focused on establishing a well-organized architecture that promotes efficient communication between modules, provides a user-friendly interface for barangay officials and residents, and ensures secure handling of data. The design also emphasized system performance, scalability, and maintainability to accommodate future updates or additional features. A well-structured system design ensured that AI-Aided BAMIS would operate smoothly, deliver accurate information, and enhance the efficiency, transparency, and responsiveness of barangay management processes.

Hierarchical Input Process Output (HIPO)

The HIPO (Hierarchical Input-Process-Output) model is a structured approach used to analyze and design systems, and it played a crucial role in defining the functions and processes of the Artificial Intelligence-Aided Barangay Abaga Management Information System with n8n Automation Agent (AI-Aided BAMIS). It clearly outlined the hierarchical levels of the system and their interactions, helping the researchers understand key processes such as online service requests, administrative approval

workflows, request tracking, and automated responses. This structured analysis helped identify potential inefficiencies, delays, and areas for improvement while supporting more efficient system operation and improved user satisfaction.

In addition, the HIPO model supported the decomposition of major system functions into smaller, manageable sub-processes, making the workflows easier to understand and implement. Each critical input, such as resident information, request details, uploaded documents, and agent queries, was aligned with the appropriate processing logic, including database transactions, approval mechanisms, and n8n-powered automation workflows.

These processes resulted in outputs such as approved documents, status updates, and automated responses. This approach enhanced system reliability and reduced errors in handling barangay services. By documenting each level of the system using HIPO diagrams, the researchers established a solid foundation for system development, testing, maintenance, and future enhancements.

Overall, the HIPO model strengthened transparency, improved coordination among system components, and helped ensure that the processes remained aligned with the objectives of efficient and automated barangay service delivery.

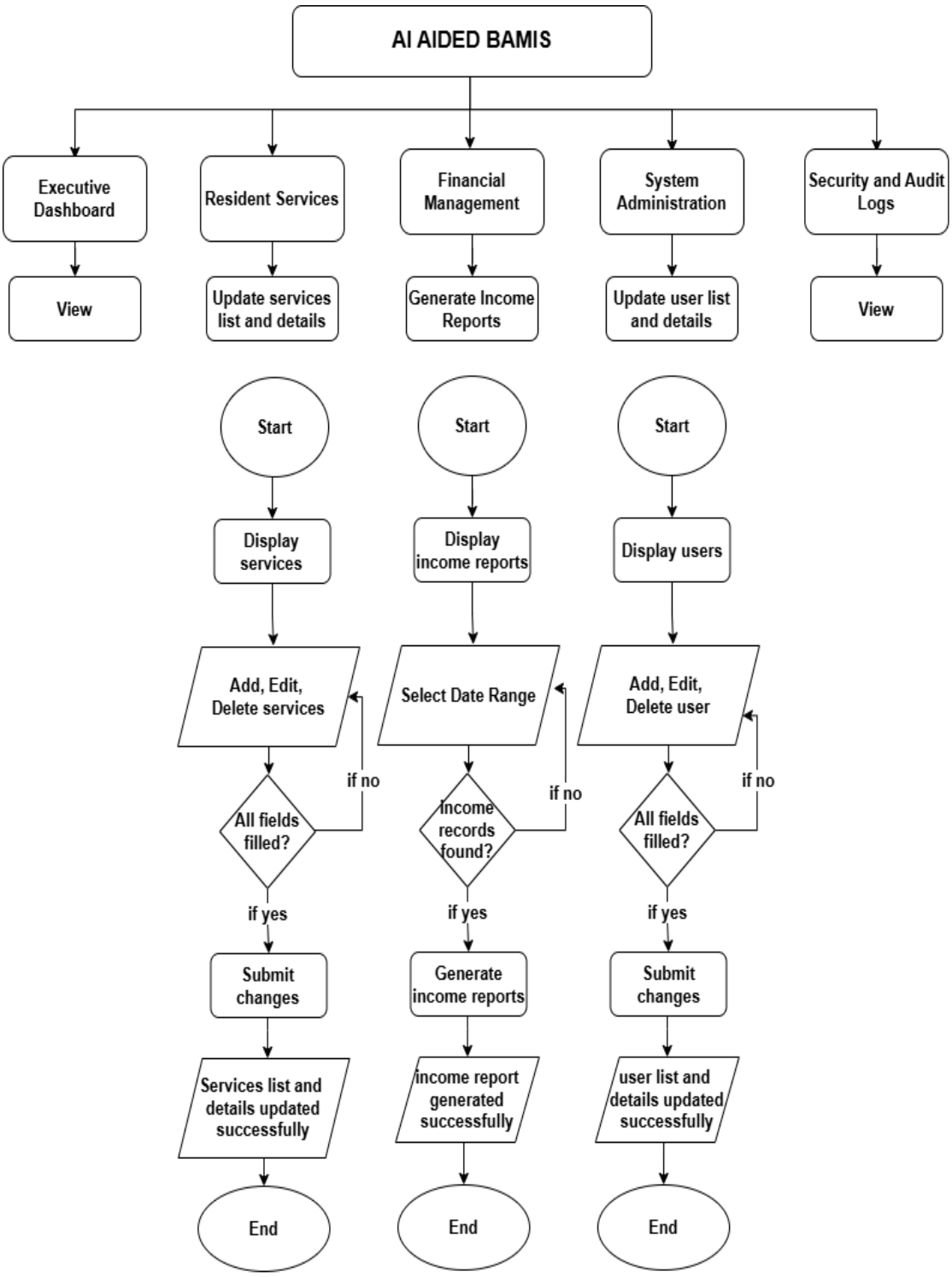


Figure 4. **Hierarchical Input Process Output (Admin) of AI Aided BAMIS**

Figure 4 presents the Hierarchical Input-Process-Output (HIPO) Diagram of the Admin Panel of the Artificial Intelligence Aided Barangay Abaga Management Information System with n8n Automation Agent. It illustrates the system's five main administrative modules: Executive Dashboard, Resident Services, Financial Management, System Administration, and Security and Audit Logs. Each module followed a structured workflow in which administrators accessed information, performed necessary actions, validated inputs, and generated corresponding outputs.

The Executive Dashboard provided administrators with an overview of important barangay information. The Resident Services module enabled administrators to add, edit, and delete service records, with all required fields validated before updates were saved. The Financial Management module processed and validated income records based on a selected date range to generate income reports. The System Administration module managed user accounts by ensuring that all user information was complete before changes were submitted. Meanwhile, the Security and Audit Logs module allowed administrators to monitor system activities and review audit records.Through these modules, the Admin Panel supported efficient management, accurate data processing, secure system administration, and transparent monitoring of barangay operations. It also contributed to improved service delivery, decision-making, and overall operational effectiveness.

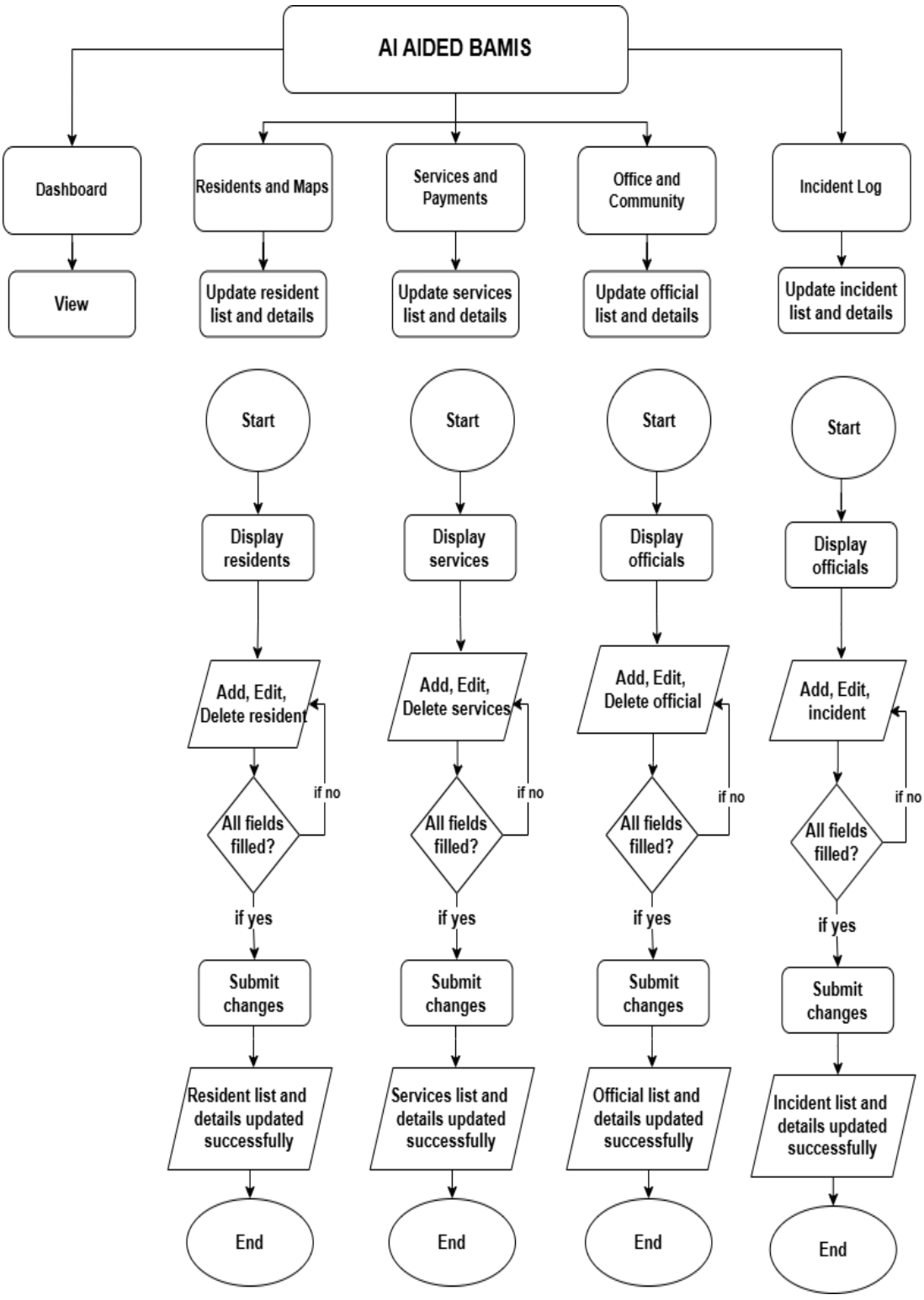


Figure 5. **Hierarchical Input Process Output (Secretary) of AI Aided BAMIS**

Figure 5 presents the Hierarchical Input-Process-Output (HIPO) Diagram of the Secretary Module of the Artificial Intelligence Aided Barangay Abaga Management Information System with n8n Automation Agent. It illustrates the main functions performed by the secretary through five modules: Dashboard, Residents and Maps, Services and Payments, Office and Community, and Incident Log. Each module followed a structured workflow in which the secretary viewed information, performed necessary actions, validated inputs, and updated records within the system.

The Dashboard served as the main interface where the secretary viewed important barangay information and system activities. The Residents and Maps module allowed the secretary to display resident records and perform add, edit, or delete operations. All required fields were validated before changes were submitted to ensure accurate resident information.

The Services and Payments module enabled the secretary to manage service records by displaying services and updating their details through validated inputs before saving changes. The Office and Community module focused on managing barangay officials, allowing the secretary to add, edit, or delete official records while ensuring that all required information was complete before submission. Meanwhile, the Incident Log module allowed the secretary to manage incident records by displaying existing reports and updating incident details. The system validated all entries before saving updates to maintain accurate and organized records.

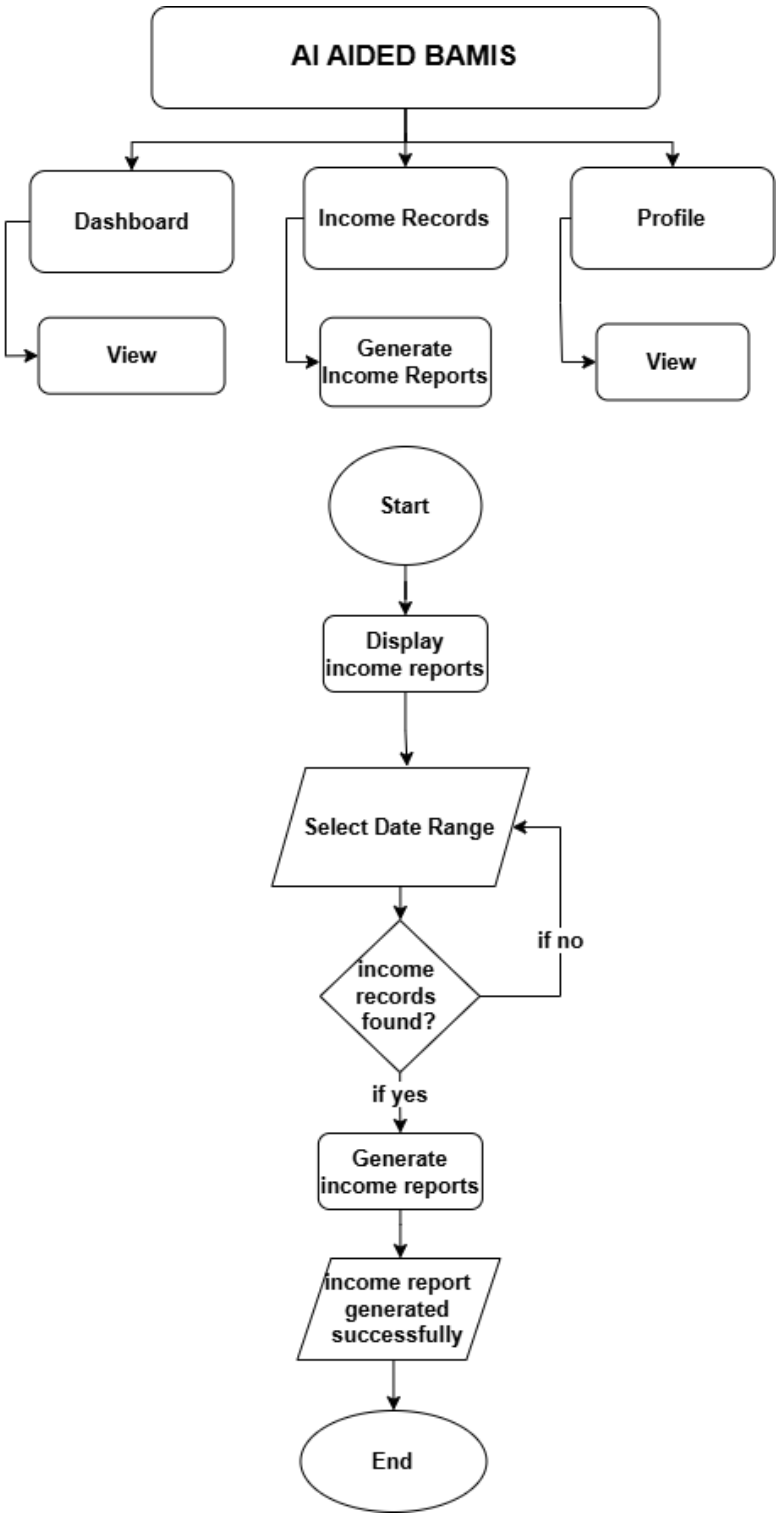


Figure 6. **Hierarchical Input Process Output (Treasurer) of AI Aided BAMIS**

Figure 6 presents the Hierarchical Input-Process-Output (HIPO) Diagram of the Treasurer Panel of the Artificial Intelligence Aided Barangay Abaga Management Information System with n8n Automation Agent. It illustrates the main functions available to the treasurer through three modules: Dashboard, Income Records, and Profile. Each module followed a structured workflow that enabled the treasurer to view information, generate reports, and manage account-related data. The Dashboard served as the main interface where the treasurer viewed financial information and accessed system functions. The Income Records module focused on generating income reports. The process began by displaying income records, after which the treasurer selected a specific date range. The system then checked whether income records existed within the selected period. If no records were found, the process returned to the date range selection stage. If records were available, the system generated the income report and displayed the output successfully. The Profile module allowed the treasurer to view personal account information and manage profile-related details within the system. Through these processes, the Treasurer Panel supported efficient monitoring of income records, accurate report generation, organized financial management, and improved transparency in barangay financial operations and reporting. It also helped ensure data accuracy, timely access to income information, and reliable documentation of financial records.

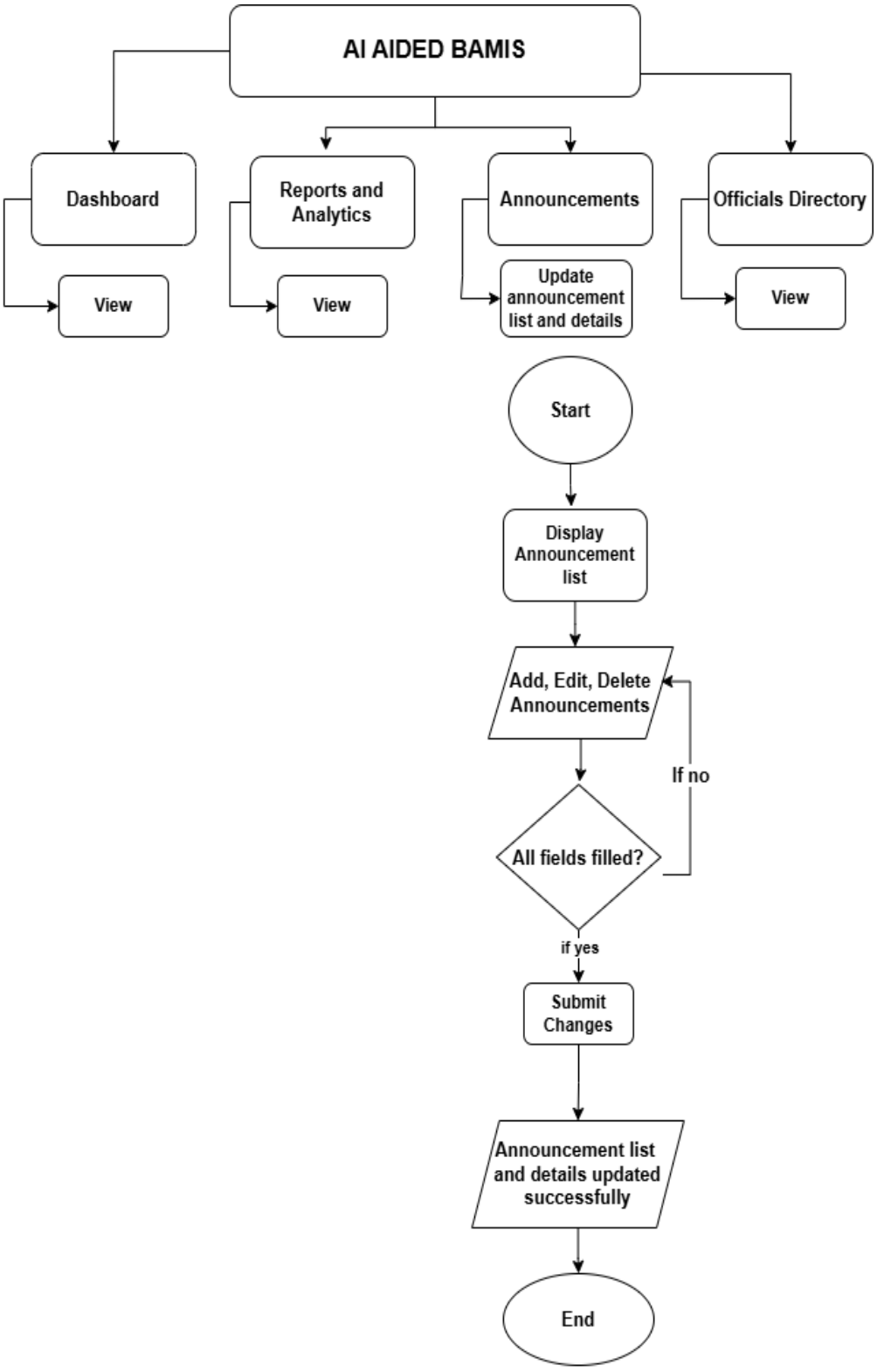


Figure 7. Hierarchical Input Process Output (Captain) of AI Aided BAMIS

Figure 7 presents the Hierarchical Input-Process-Output (HIPO) Diagram of the Captain Panel of the Artificial Intelligence Aided Barangay Abaga Management Information System with n8n Automation Agent. It illustrates the primary functions available to the Barangay Captain through four modules: Dashboard, Reports and Analytics, Announcements, and Officials Directory. Each module followed a structured workflow that allowed the Captain to access information, perform administrative actions, validate inputs, and generate corresponding outputs. The Dashboard served as the main interface where the Captain viewed important barangay information. The Reports and Analytics module allowed the Captain to view reports and analytical data that supported the monitoring and evaluation of barangay activities. The Announcements module enabled the Captain to manage community announcements by displaying existing announcement records and allowing the addition, editing, or deletion of announcements. Before any changes were saved, the system validated whether all required fields had been completed. Once validated, the updates were submitted, and the announcement list was successfully updated. Meanwhile, the Officials Directory module allowed the Captain to view information about barangay officials. Through these modules, the Captain Panel supported effective monitoring of barangay activities, efficient management of announcements, reliable access to reports and official records, and improved communication and transparency within the barangay community.

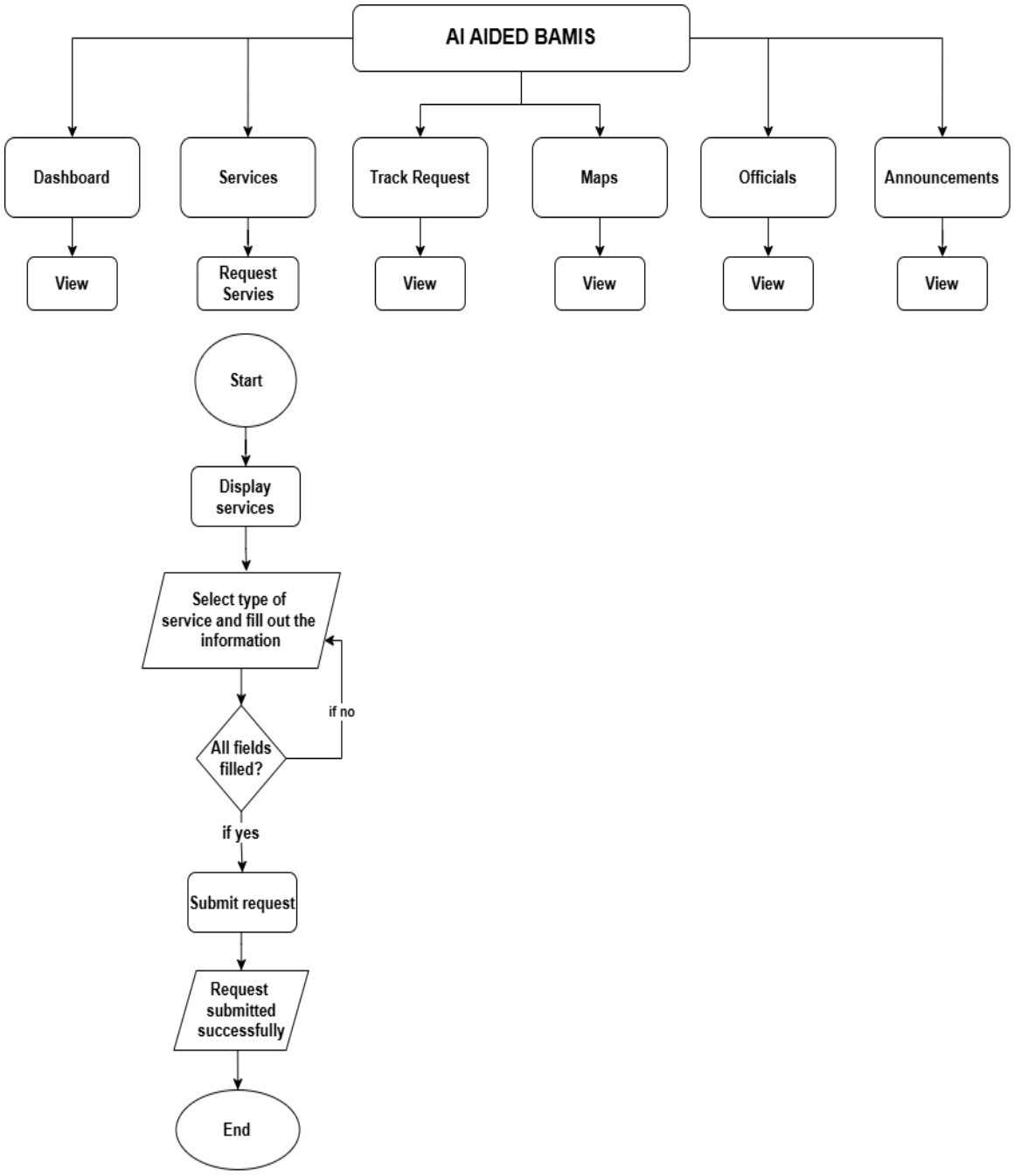


Figure 8. **Hierarchical Input Process Output (Resident) of AI Aided BAMIS**

Figure 8 presents the HIPO (Hierarchical Input-Process-Output) flow of the Resident Panel in the Artificial Intelligence–Aided Barangay Abaga Management Information System with n8n Automation Agent. It illustrates how residents interacted with the system through six modules: Dashboard, Request Services, Track Request, Maps, Officials, and Announcements, following a structured flow of accessing information, performing actions, validating inputs, and receiving outputs.

In the Dashboard, residents viewed an overview of the system and selected the feature they wanted to use, serving as the main entry point for transactions. The Request Services module handled service applications. Residents selected a service, entered the required details, and submitted requests after the system checked whether the information was complete and valid. Incorrect entries were returned for correction, ensuring proper and accurate submissions.

In the Track Request module, residents monitored the status of their submitted requests through filtered views, allowing them to stay updated without having to visit the barangay office. The remaining modules provided access to relevant information. The Maps module displayed location data, the Officials module displayed barangay personnel information, and the Announcements module provided updates and notices. These modules supported easier access to public information and improved community awareness.

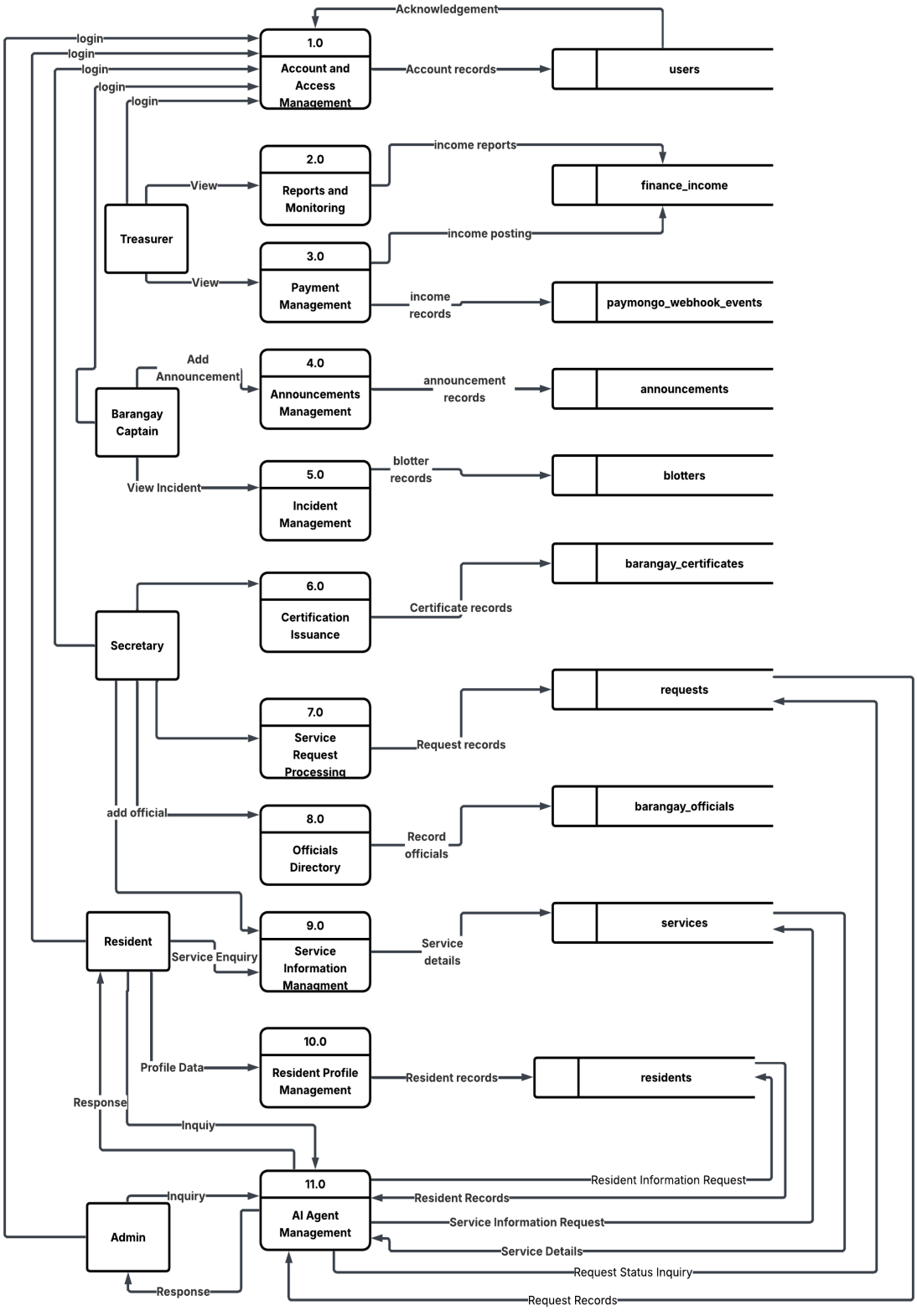


Figure 9. **Data Flow Diagram of AI Aided BAMIS**

Figure 9 presents the Data Flow Diagram (DFD) of the Artificial Intelligence Aided Barangay Abaga Management Information System with n8n Automation Agent. It illustrates how data flowed between system users, processes, and data stores to support barangay operations. The primary users of the system were the Resident, Secretary, Treasurer, Barangay Captain, and Admin, with each user interacting with specific modules based on their assigned responsibilities. The diagram also showed how information was processed, stored, retrieved, and shared across different components of the system. The process began with Account and Access Management, where all users logged in and accessed the system. User account information was stored in the users data store, which maintained account records and authentication details. Once authenticated, users were able to access their assigned modules and perform their designated functions.

The Treasurer interacted with Reports and Monitoring and Payment Management. Through Reports and Monitoring, the Treasurer viewed income reports generated from financial records stored in the finance_income data store. In Payment Management, income postings and payment records were processed and stored in the paymongo_webhook_events data store, ensuring proper recording and monitoring of barangay income transactions. The Barangay Captain managed information through Announcements Management and Incident Management. Announcement records were created and maintained within

the announcements data store, allowing important barangay information to be disseminated. The Captain also reviewed incident or blotter records through Incident Management, with the corresponding data stored in the blotters data store for documentation and monitoring purposes. The Secretary performed several administrative functions, including Certification Issuance, Service Request Processing, and Officials Directory. Certificate records generated through Certification Issuance were stored in the barangay_certificates data store. Service requests submitted by residents were processed and recorded in the requests data store.

The Resident interacted with Service Information Management and Resident Profile Management. Through these modules, residents were able to view service information, submit inquiries, and update personal details. Service information was retrieved from the services data store, while resident information was maintained in the residents data store to ensure accurate and updated resident records. The system also included AI Agent Management, which served as the intelligent component of AI-Aided BAMIS. The AI Agent received inquiries from residents and administrators, retrieved relevant information from the residents, services, and requests data stores, and provided appropriate responses. It processed resident information requests, service information requests, and request status inquiries, enabling faster access to information and improving user support within the system.

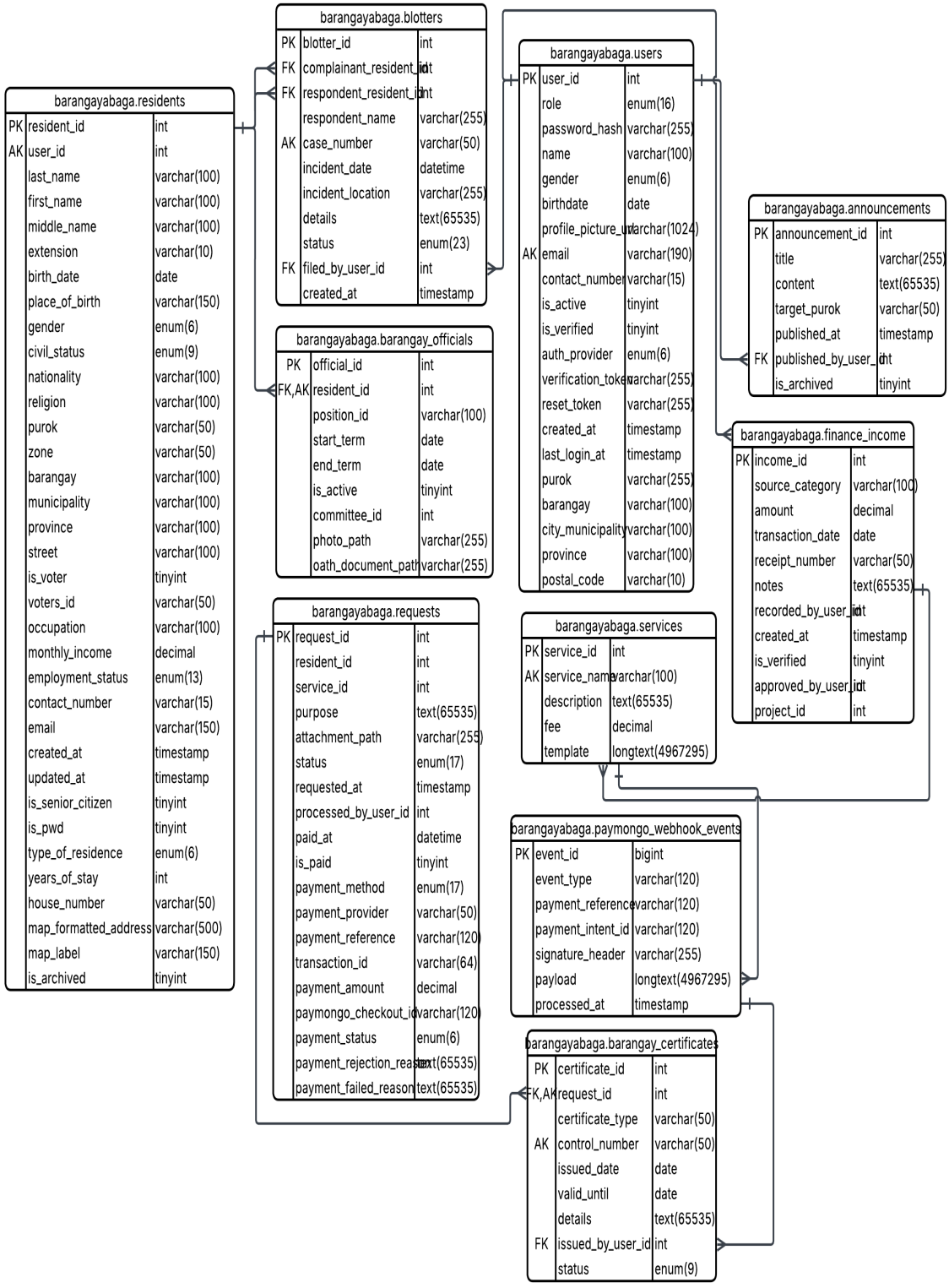


Figure 10. **Entity Relationship Diagram of AI Aided BAMIS**

In Figure 10, the Entity Relationship Diagram (ERD) of the AI-Aided BAMIS with n8n Automation Agent presents the overall structure of the system database, highlighting how key entities were interconnected to support efficient barangay operations and automation. At the core was the residents entity, which stored comprehensive personal and demographic information and served as the central reference for transactions. Residents were linked to requests for services and blotters for incident reports, ensuring that records were accurately associated with individuals.

The users entity managed system access and administrative roles, allowing authorized personnel to handle announcements, requests, certificates, blotters, and financial records. Services offered by the barangay were defined in the services entity and connected to requests, where residents applied for specific services. Approved requests resulted in the generation of records in the barangay_certificates entity, ensuring traceability from application to document issuance. Incident management was handled through the blotters entity, which recorded complaints and linked involved residents and responsible users. Administrative roles were maintained in the barangay_officials entity, connecting residents to their official positions and terms. Communication was supported by the announcements entity, which enabled targeted information dissemination, while financial transactions were recorded in the finance_income entity to promote transparency.

Furthermore, the integration of the paymongo_webhook_events entity enabled real-time tracking of online payments and supported automated processing through the n8n automation agent. Overall, the ERD reflected a structured, resident-centered database design that supported data consistency, accountability, and efficient management of services, records, and operations within the system. It also supported scalable automation, improved data synchronization, and enhanced system reliability across the interconnected modules.

Implementation Phase. The implementation phase of the Artificial Intelligence-Aided Barangay Abaga Management Information System with n8n Automation Agent focused on transforming the system design into a functional, intelligent, and automated platform. The researchers developed the system by coding software components, integrating AI capabilities, and utilizing n8n to automate workflows and enhance system functionality. The implementation included system integration, AI-assisted automation, and thorough testing to ensure reliability, accuracy, and user-friendliness across the different modules and user roles. The phase resulted in a functional solution in which AI and automated workflows supported residents, barangay officials, and administrators in managing service requests, certificates, announcements, financial transactions, and system information efficiently and securely. The implemented features helped reduce manual

workload and improved operational efficiency while supporting system stability, maintainability, and potential future enhancements.

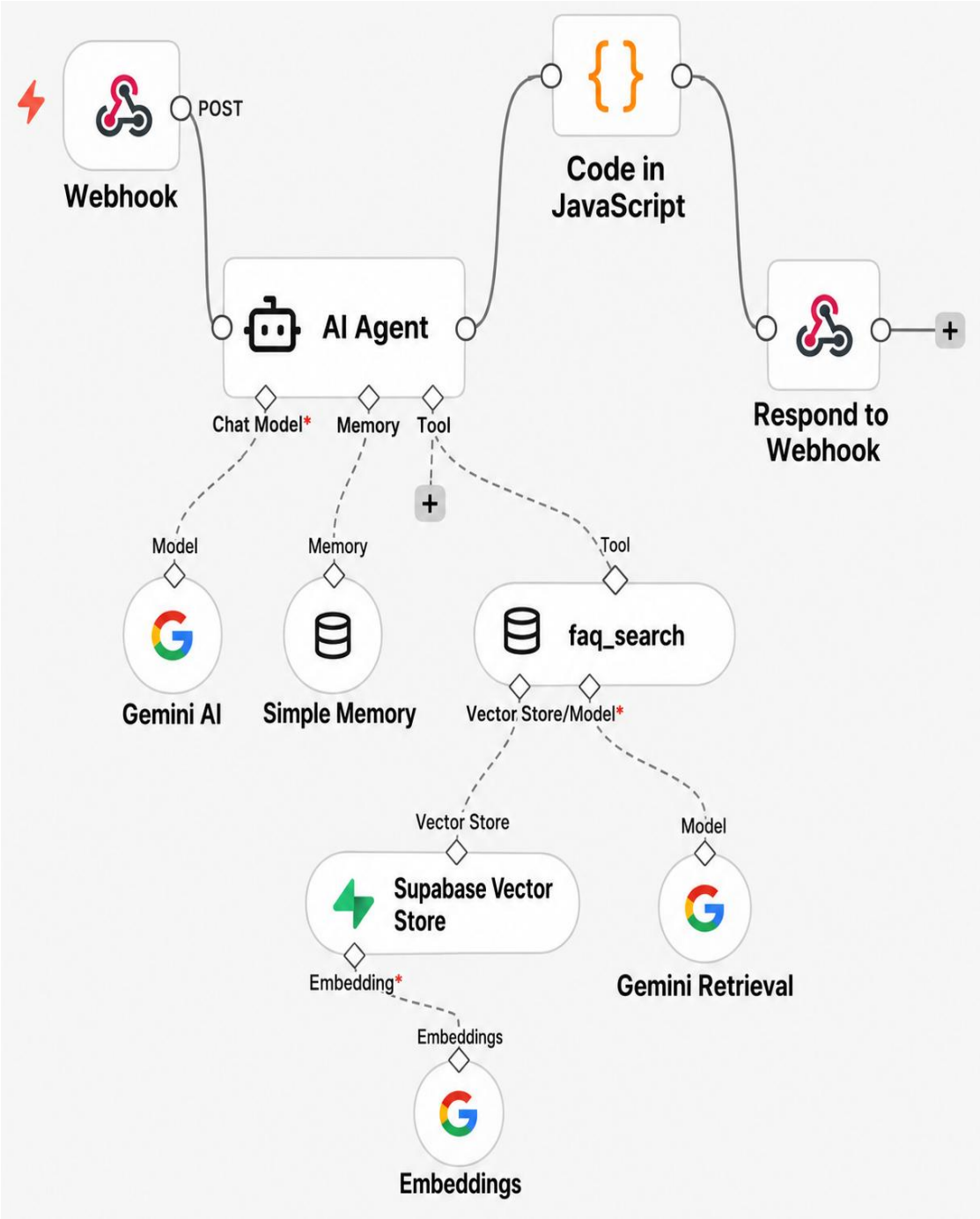


Figure 11. **Resident Agent Workflow of AI Aided BAMIS**

In Figure 11, Resident Agent Workflow, the diagram illustrates how the system used a Retrieval-Augmented Generation (RAG) approach within n8n to process user queries intelligently. The workflow began with a Webhook that received the user's message and session ID, which were then passed to the AI Agent. Acting as the core component of the system, the AI Agent followed predefined instructions that prioritized retrieving accurate information from a connected knowledge base rather than relying solely on its internal knowledge.

To accomplish this, the AI Agent utilized the faq_search tool, which connected to the Supabase Vector Store where the BAMIS knowledge base was stored. The user's query was transformed into embeddings using Google Gemini, enabling semantic searches within the database to retrieve the most relevant information.

The retrieved data snippets were then used by the AI model to generate a precise and context-aware response, while the Simple Memory component maintained continuity by keeping track of recent interactions. Finally, the generated response was processed through a JavaScript node, which cleaned and formatted the output into a structured JSON response. The resulting response was then sent back to the user through the Respond to Webhook node.

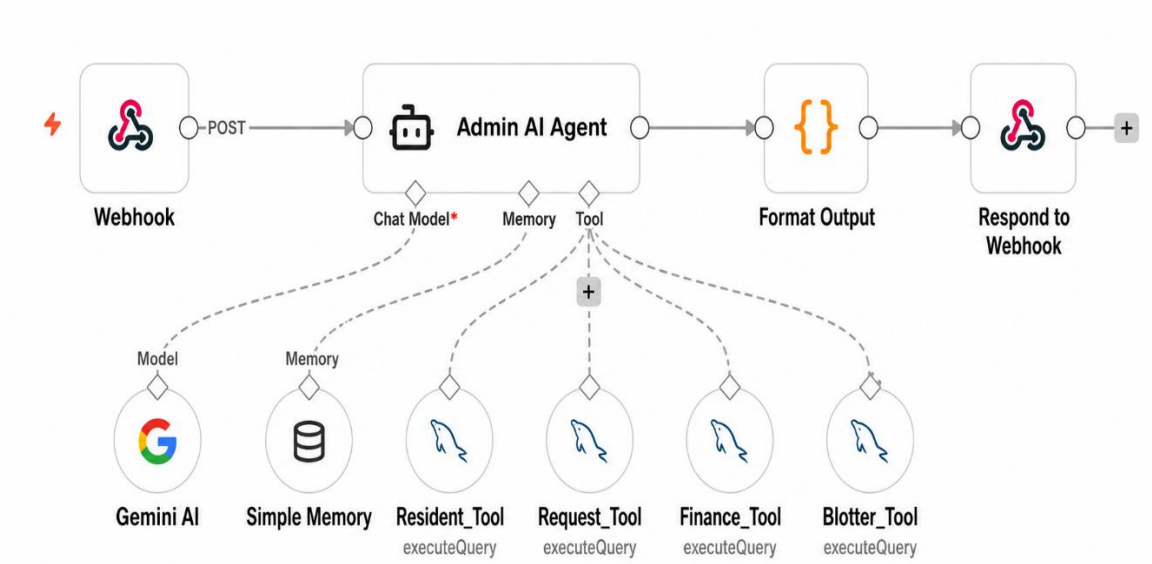


Figure 12. **Admin Agent Workflow of AI Aided BAMIS**

In Figure 12, Admin Agent Workflow, the system architecture illustrates a streamlined and automated process flow for handling administrative queries within the Barangay Abaga Administrative AI Agent. The workflow began with a Webhook node, which served as the primary entry point for incoming HTTP POST requests from the frontend application. These requests contained the user's query and session identifier, which were essential for tracking and contextualizing interactions. The request was then passed to the Admin AI Agent, the core decision-making component responsible for interpreting the query using natural language processing and determining the appropriate course of action.

The Admin AI Agent was integrated with multiple supporting components, including the Gemini AI model for language understanding and

several specialized tools such as Resident_Tool, Request_Tool, Finance_Tool, and Blotter_Tool. These tools enabled the agent to execute specific SQL queries based on the user's intent, allowing the dynamic retrieval of relevant data from the database. Additionally, the Simple Memory module maintained conversational continuity by storing recent interactions associated with the session ID. This enabled the system to handle follow-up queries with contextual awareness. This modular integration demonstrated the system's ability to combine AI-assisted processing with structured data access.

After the AI Agent processed the request and retrieved or generated the appropriate response, the output was forwarded to the Format Output node. This stage ensured that all responses conformed to a standardized JSON structure, improving consistency and reducing potential frontend parsing errors. Finally, the Respond to Webhook node sent the processed and formatted response back to the originating client, thereby completing the request-response cycle.

Overall, the workflow demonstrated an efficient, secure, and scalable architecture that leveraged automation and AI-assisted decision-making to support barangay-level administrative operations.

Calendar of Activities

The development of the Artificial Intelligence-Aided Barangay Abaga Management Information System with n8n Automation Agent followed a structured and systematic timeline, as presented in the project's calendar of activities. The schedule outlined the sequence of development phases from December 2025 to April 2026, ensuring that each stage of the system development process was properly organized and completed within the designated timeframe. This organized approach enabled the researchers to effectively manage resources, coordinate tasks, and monitor progress while maintaining alignment with the primary objective of improving the efficiency of administrative services and records management in Barangay Abaga.

The project began with the Planning Phase, which was conducted during the early weeks of December 2025. During this stage, the researchers established the overall direction of the project by defining the system's objectives, scope, and expected outcomes. The planning process also involved identifying the primary stakeholders of the system, including barangay officials, administrative staff, and residents who would utilize the platform. In addition, the researchers evaluated the technical and operational feasibility of implementing a digital management system within the barangay to determine whether the proposed solution could effectively support its administrative processes.

Following the planning stage, the Analysis Phase was conducted during the early part of January 2026. This phase focused on identifying and examining the specific requirements necessary for the system to function effectively. The researchers gathered information through consultations and discussions with barangay officials and personnel to understand existing workflows and operational challenges. Essential system requirements were identified, including resident profiling, document request processing, user authentication, role-based access control, and request tracking. Particular emphasis was placed on maintaining data accuracy, security, and privacy to support appropriate data management practices.

The System Design Phase followed the completion of the analysis stage and extended from late January to February 2026. During this phase, the researchers translated the identified requirements into a detailed system architecture and technical framework. This process included designing the database structure that stored and managed barangay records such as resident information, household data, transaction records, and service requests. System diagrams and interface prototypes were also developed to guide the development process and ensure that the platform’s structure aligned with the operational workflow of the barangay. The design stage ensured that both the technical components and user interface were organized, efficient, and user-friendly.

Parallel to the design and development preparations, the Research Proposal Preparation Stage was conducted, during which the researchers compiled the gathered data, conceptual frameworks, and methodological approaches into a formal research document. This documentation provided a comprehensive explanation of the system's purpose, development methodology, and expected contributions to barangay governance and service delivery.

The Implementation Phase began during the first week of March 2026 and continued until early April 2026. This stage involved the actual development of the system through coding and integration of its various components. Core system functionalities were developed, including modules for resident management, document processing, and request monitoring. During this stage, the researchers ensured that the system modules were properly integrated and that the platform operated as intended within a web-based environment.

Following the implementation stage, the Testing Phase was conducted to verify the functionality, reliability, and usability of the system. The researchers performed a series of tests to ensure that the system operated accurately and efficiently. These tests included checking system modules, verifying database interactions, and evaluating the user interface to assess ease of use. The purpose of this phase was to identify and resolve

issues that could affect system performance before the system was introduced for operational use.

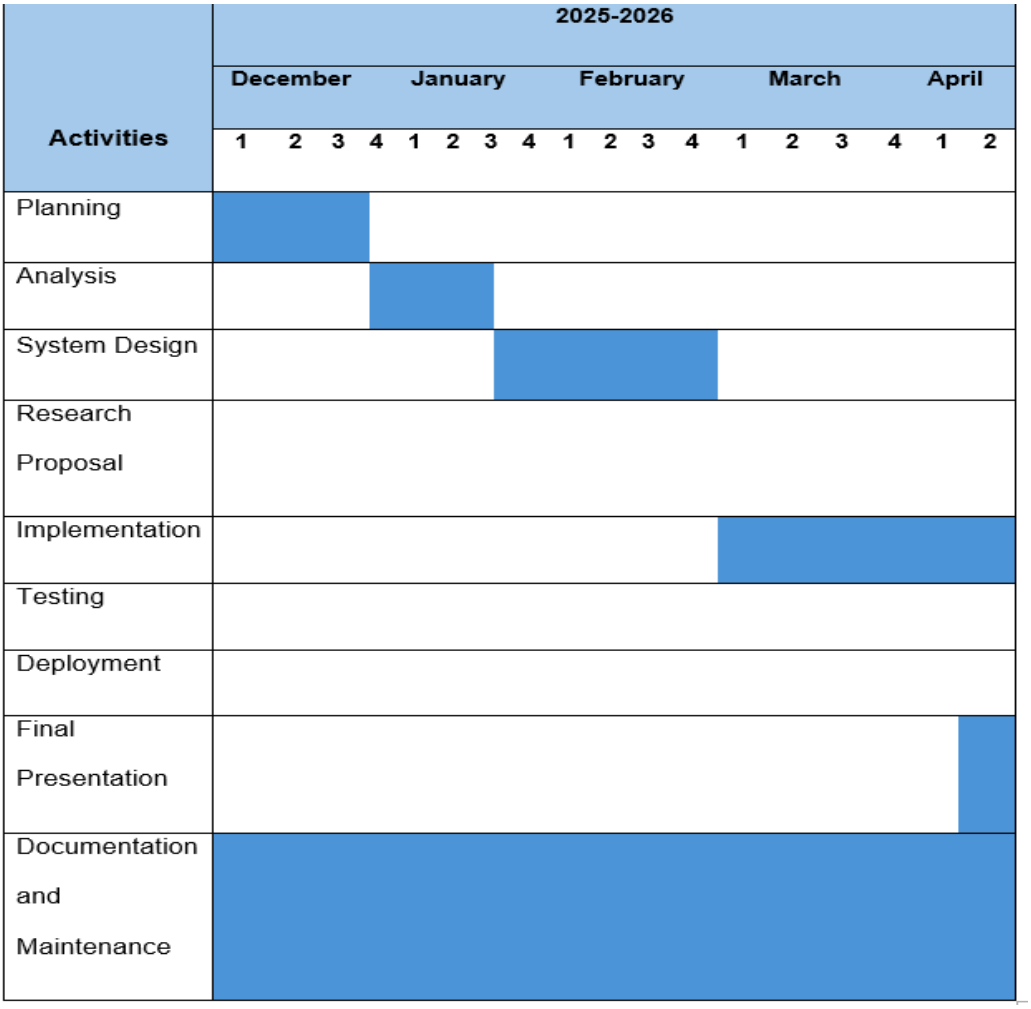
After successful testing, the project proceeded to the Deployment Phase, during which the system was prepared for operational use within the barangay environment. This stage involved configuring the system for its intended operational setting and ensuring that barangay officials and personnel were capable of utilizing its features effectively. The deployment phase marked the transition of the system from development to practical implementation.

The final stage of the project was the Final Presentation, which was scheduled for April 2026. During this phase, the researchers presented and demonstrated the completed system, highlighting its features, functionalities, and contributions to improving barangay administrative operations. The presentation served as the culmination of the project and showcased how the system was designed to enhance efficiency, transparency, and service delivery within Barangay Abaga.

Throughout the project duration, from December 2025 to April 2026, documentation and maintenance activities were conducted continuously. The researchers recorded the technical details of the system, including development procedures, database structures, and system configurations. Continuous documentation helped ensure that the system remained

maintainable, scalable, and adaptable for potential future enhancements in response to the technological and administrative needs of the barangay.

Table 3. **Calendar of Activities from December 2025 to April 2026**



Testing. During the testing phase of the Artificial Intelligence-Aided Barangay Abaga Management Information System with n8n Automation Agent, the primary objective was to ensure the system's functionality, reliability, and efficiency in managing barangay operations, resident-related

transactions, and AI-assisted processes. This stage involved a comprehensive evaluation of all system components and applied testing methodologies such as functionality testing, performance testing, and security testing.

Each module, including resident profiling, document request management, user authentication, and automated workflow processes powered by n8n, was carefully examined to ensure accurate data processing and stable performance under various conditions. User Acceptance Testing (UAT) was also conducted as an important part of the evaluation, allowing barangay officials, staff, and selected residents to assess how well the system met their operational and AI-assisted service needs.

The testing phase demonstrated that the AI-Aided BAMIS was functional, reliable, and capable of supporting barangay administrative operations. The results indicated that the system could contribute to improved administrative efficiency, transparency, and responsiveness within Barangay Abaga.

Deployment. In the final deployment phase of the Artificial Intelligence-Aided Barangay Abaga Management Information System with n8n Automation Agent, all essential infrastructure components were integrated and configured to support smooth and secure operation in the intended environment. This phase involved transferring the finalized database and

application files from the development environment to the production server, along with conducting thorough compatibility and security checks.

Prior to deployment, barangay officials and staff were provided with orientation and training sessions to familiarize them with the system's functions, workflows, and AI-assisted features, including n8n-powered automated tasks. The deployment process also included a planned rollout strategy to support the transition from traditional manual record-keeping to a digital and automated information management system.

Emphasis was placed on data accuracy, migration, and security throughout the transition process. Feedback from initial users was also collected to identify and address minor technical issues. These activities helped ensure that the system operated properly and effectively supported the intended barangay services.

Maintenance. The maintenance phase of the Artificial Intelligence-Aided Barangay Abaga Management Information System with n8n Automation Agent played an important role in sustaining its functionality, security, and adaptability to future technological and administrative needs. This stage involved continuous monitoring of system performance, periodic updates, and database maintenance to ensure consistent operation, data integrity, and proper functioning of AI-assisted processes. Regular system audits and feedback assessments were conducted to identify areas for improvement, while new features and optimizations were introduced when necessary to

address evolving barangay requirements. Maintenance activities included fixing bugs, improving the user interface for better accessibility, updating software dependencies, and maintaining workflow automation powered by n8n. This proactive approach helped ensure that the AI-Aided BAMIS remained a reliable, efficient, and user-friendly platform that supported effective governance, data management, and citizen service delivery in Barangay Abaga.

Research Environment

The research was conducted in Barangay Abaga, located in the Municipality of Lala, Lanao del Norte, Philippines. Barangay Abaga was a progressive and rapidly developing community situated along the national highway that connected key cities such as Iligan and Ozamiz. It served as one of Lala's vital residential and commercial zones, providing access to nearby schools, markets, and subdivisions such as La Virginia Homes. The barangay's strategic location contributed to its growing importance in local governance and digital transformation initiatives. Surrounded by a combination of agricultural lands and emerging establishments, Abaga represented a balance between rural and semi-urban development in Northern Mindanao. The area experienced a tropical climate characterized by alternating wet and dry seasons, which influenced both its agricultural productivity and the daily activities of its residents.

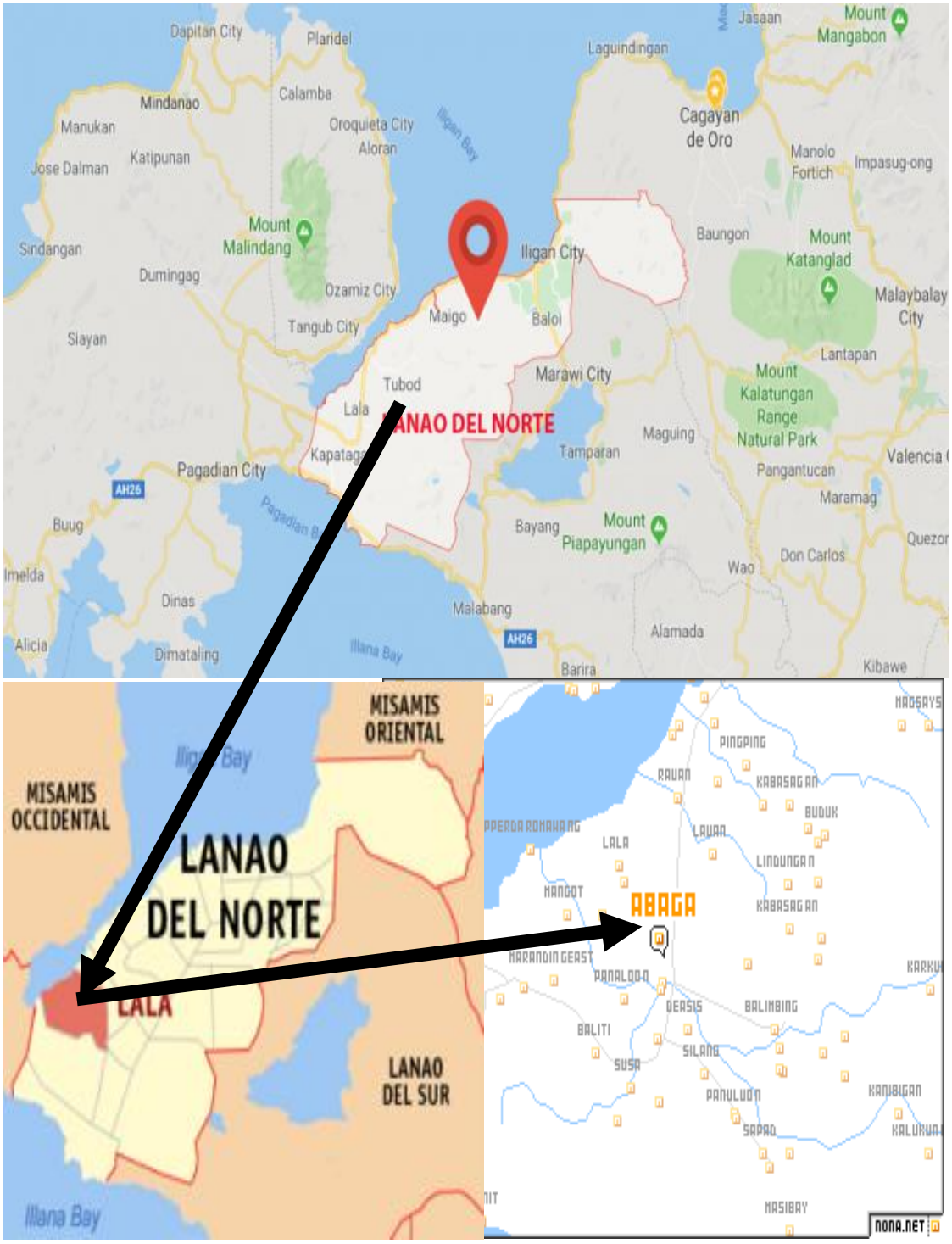


Figure 13. **The Locale of the Study**

Research Respondents

The participants in this study included the Barangay Captain, Secretary, Treasurer, Barangay Staff, and selected residents of Barangay Abaga, each of whom played a vital role in the overall management and operation of the barangay. These respondents were carefully selected based on their responsibilities, familiarity with existing administrative processes, and relevance to the implementation and evaluation of the Barangay Abaga Management Information System (BAMIS).

The Barangay Captain provided leadership and oversight, ensuring that the system was aligned with the barangay's goals of improving governance and service delivery. The Barangay Secretary provided essential insights regarding documentation, record-keeping, and the issuance of certificates, while the Treasurer provided input related to financial monitoring and transparency. The Barangay Staff served as system users responsible for managing records, handling requests, and assisting residents, making their feedback crucial in evaluating the system's usability and functionality.

Meanwhile, selected residents represented the intended beneficiaries of the system and provided their experiences and perspectives regarding the accessibility, convenience, and satisfaction associated with digitalized barangay services.

The researchers gathered feedback from these respondents to evaluate the system's performance, reliability, and overall contribution to administrative operations and public service delivery. Their insights were analyzed to identify strengths, areas for improvement, and opportunities for future enhancements. Through this collaborative approach, the study aimed to ensure that the Barangay Abaga Management Information System effectively met the operational needs of the barangay and promoted a more efficient, transparent, and citizen-centered governance framework.

Data Gathering Procedures

The data gathering process for the Artificial Intelligence-Aided Barangay Abaga Management Information System with n8n Automation Agent was conducted to collect relevant information and user requirements needed for the development of the system. The researchers first secured approval from the research adviser and obtained permission from the officials of Barangay Abaga. Formal requests were submitted, and data collection was carried out only after the necessary approvals from the barangay office and relevant stakeholders were granted.

The researchers gathered data through interviews, observations, and a review of the existing manual processes in the barangay, particularly those related to service requests, record management, and administrative workflows. The information collected served as the basis for identifying

system requirements and designing appropriate system functionalities that aligned with the operational processes of Barangay Abaga.

Statistical Treatment of the Data

The responses gathered from the participants were analyzed to ensure accurate interpretation of the data and to support the findings of the study. The following statistical tools were used:

Frequency and Percentage. Frequency and percentage were used to determine the number and proportion of responses for each category and to describe the distribution of answers from the respondents. These measures helped identify common patterns, preferences, and opinions related to the system.

Frequency referred to the number of occurrences of a specific response, while percentage represented the proportion of each response relative to the total number of responses.

The percentage was computed using the following formula:

$$Percentage = \left(\frac{frequency}{total} number\ of\ values\right) \times 100$$

Where:

P = Percentage

f = Frequency of the response

N = Total number of respondents

100 = Constant multiplier

Weighted Mean. The weighted mean was used to determine the average response of the participants based on the assigned weights for each response, particularly for survey items that used rating scales. This statistical measure provided a more appropriate representation of the participants' overall responses by considering the corresponding weight assigned to each rating.

The weighted mean was computed using the following formula:

$$Weighted\ Mean = \frac{\sum(w_i\ x_i)}{\sum w_i}$$

Where:

WM = Weighted Mean

f = Frequency of each response

x = Weight or numerical value assigned to each response

Σfx = Sum of the products of frequency and weight

N = Total number of responses

The resulting weighted mean was used to determine the overall assessment of the participants regarding the system based on the criteria included in the evaluation instrument.

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